

AV Fundamentals for Educators

Resource Appendix and Production Companion Guide

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About This Document

This Resource Appendix and Production Companion is a reference supplement to the *AV Fundamentals for Educators* textbook package, co-published by the International Society for Technology in Education (ISTE) and AVIXA (the Audiovisual and Integrated Experience Association). It serves three primary audiences: (1) educators and instructors using the textbook in classroom or professional development settings; (2) institutions and AV coordinators planning, procuring, and managing educational AV systems; and (3) the editorial and production team responsible for developing, reviewing, and delivering the complete textbook package. All specifications, templates, and guidelines in this document are vendor-neutral unless otherwise noted.

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PART ONE: EDUCATOR RESOURCE APPENDIX

Part One provides ready-to-use reference templates, planning tools, glossary definitions, and procurement guidance for educators, AV coordinators, campus technology directors, and institutions adopting the *AV Fundamentals for Educators* curriculum. All templates are vendor-neutral and designed for direct use in educational planning, procurement, and program delivery contexts.

Section A: Bill of Materials (BOM) Templates

The following Bill of Materials templates are designed to guide educational institutions through the equipment planning and budgeting process for three common AV deployment scenarios. All specifications are performance-based and intentionally vendor-neutral; no brand names or model numbers appear in these templates. Priority designations use a three-tier system: **Must** (required for basic functional operation), **Should** (strongly recommended for effective instruction), and **Nice** (beneficial but deferrable if budget is constrained). Cost ranges are illustrative estimates only and should be validated against current market pricing at time of procurement.

Important: Cost Range Disclaimer

All cost estimates in the BOM templates reflect general market ranges as of 2025–2026. Actual pricing will vary based on region, quantity, cooperative purchasing agreements, installation complexity, and market conditions. Always obtain a minimum of three competitive quotes or proposals before finalizing a project budget. Institutions eligible for E-Rate funding should consult their E-Rate consultant before finalizing specifications.

Template A-1: Basic Classroom AV Bill of Materials

Scenario: Standard instructional classroom, 20–35 students, single-space configuration, primary use for in-room instruction with occasional video conferencing via laptop.

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
Interactive Flat Panel Display	Minimum 75-inch diagonal, 4K UHD (3840×2160) resolution, minimum 400 nits brightness, touch-enabled (20-point multi-touch), anti-glare coating, built-in media player, front-facing speakers (minimum 2×20W), USB-C and HDMI inputs, wireless casting support	1	\$3,500–\$6,500	\$3,500–\$6,500	Must	Large text display, touch interaction, captioning display surface	Mount height must comply with ADA (operational controls ≤54" AFF for side reach; ≤48" for forward reach)
Document Camera	Minimum 1080p (1920×1080) output resolution, USB 3.0 connectivity, flexible arm with	1	\$200–\$500	\$200–\$500	Should	Supports visual accommodation for close-up material display	Verify USB driver compatibility with district's standard OS image

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	360° rotation, auto-focus, built-in LED illumination, plug-and-play compatibility with major OS platforms						
Ceiling Speaker System	2-speaker distributed audio configuration, 70V/100 V constant-voltage system, minimum 8-inch driver size per speaker, frequency response 80Hz–16kHz (±3dB), ceiling-mount back-can enclosures, UL-listed	1 system (2 spkrs)	\$400–\$900	\$400–\$900	Must	Provides even sound distribution for students with hearing challenges; required for assistive listening system integration	Placement per ANSI S12.60 guidelines; coordinate with ceiling tile layout
Digital Signal Processor / Audio Amplifier	Integrated DSP with built-in acoustic	1	\$800–\$2,000	\$800–\$2,000	Must	AEC critical for equitable	Select unit with dante or AoIP capability

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	echo cancellation (AEC), automatic gain control (AGC), noise suppression, minimum 2-channel amplification at 70V, network control capability (IP-based), RJ-45 and USB audio inputs					participation in hybrid/video conferencing scenarios	y for future expansion
Lapel/Lavalier Wireless Microphone System	UHF or DECT wireless operation, single-channel receiver with diversity antennas, rechargeable bodypack transmitter, cardioid lavalier capsule included, minimum 50 ft	1	\$500–\$1,200	\$500–\$1,200	Must	Amplified instructor voice supports students with hearing difficulties and assistive listening system users	Verify channel coordination with adjacent rooms; avoid 600 MHz band (FCC reallocation)

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	operating range in open space, USB or XLR output, frequency coordination capability						
HDMI Matrix Switcher	Minimum 4 inputs × 2 outputs, 4K/60Hz (4:4:4) pass-through, HDCP 2.2 compliant, EDID management, RS-232 and IP control, auto-switching capability, 1RU rackmount form factor	1	\$600–\$1,500	\$600–\$1,500	Must	Supports multi-source display including captioning systems and accessible content players	Confirm HDCP compliance with all connected displays before specifying
Wireless Presentation System	Dual-band Wi-Fi (2.4GHz and 5GHz) operation, content sharing	1	\$500–\$1,500	\$500–\$1,500	Should	Reduces physical barriers to content sharing; allows BYOD from accessible	Requires AV VLAN configuration with IT; test with institutional Wi-Fi policy

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	from Windows , macOS, iOS, and Android, AirPlay and Miracast compatible, no-app option available , maximum 1080p streaming resolution, IT-manageable via web interface					e devices	
CAT6A Structured Cabling	CAT6A (10GbE-rated), plenum-rated (CMP) where required by code, 23 AWG, shielded (F/UTP) recommended in EMI-prone environments, installed with keystone jacks, labeled per TIA-606,	Per drop (est. 4-6 drops per room)	\$150-\$350 per drop (installed)	\$600-\$2,100	Must	Required for networked AV and assistive technology device connectivity	Include drops for: IT network, control processor, DSP, display, scheduling panel

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	tested to TIA-568-C.2 standard						
HDMI Cables (Active, 4K-rated)	2-meter length, active HDMI 2.0b, 4K/60Hz (18Gbps) rated, HDCP 2.2 compliant, strain-relieved connectors, CL2-rated jacket for in-wall runs	6	\$25-\$60 each	\$150-\$360	Must	N/A	Quantity based on number of sources; over-order by 20% for spares
USB-C to HDMI Adapter Kit	USB-C to HDMI 2.0 adapter, 4K/60Hz support, DisplayPort Alt Mode compatible, universal compatibility (no driver required), compact form factor	4	\$30-\$60 each	\$120-\$240	Should	Enables BYOD connectivity from accessible personal devices	Store in labeled drawer at instructor station; include in faculty orientation
AV Control Touchpanel	Minimum 7-inch color touchscreen, IP-	1	\$600-\$1,500	\$600-\$1,500	Should	Must be mounted at ADA-compliant height	UI design must be reviewed for

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	based (PoE+ powered), capacitive glass surface, wall-mount gang box compatible, programmable by control processor, ADA-compliant height installation required					(≤48" AFF forward reach); font size minimum 14pt on UI; high-contrast UI required	WCAG 2.1 AA contrast compliance
Power Conditioning / UPS	Minimum 1500VA / 900W output capacity, pure sine wave output, automatic voltage regulation (AVR), minimum 6 protected outlets, network interface for remote monitoring, UL-listed,	1	\$400-\$900	\$400-\$900	Must	Ensures uninterrupted operation of accessibility systems (hearing loop, captioning) during brief power events	Size UPS to actual connected load; battery replacement cycle approximately 3-5 years

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	minimum 10-minute runtime at 50% load						
Assistive Listening System (FM)	FCC Part 15-compliant FM assistive listening system, minimum 4 receiver units included, single transmitter, 72–76 MHz frequency range, neck loop accessory included per receiver, carrying case, dedicated charging station	1 system (4 receivers)	\$500–\$1,200	\$500–\$1,200	Must	Required by ADA for places of public accommodation with audio systems; ANSI/ICC A117.1 reference	Number of receivers per ADA: minimum 2 or 25% of capacity (whichever is greater); 4-receiver kit meets small classroom requirement
Hearing Loop (Room Perimeter Induction Loop)	Room perimeter copper wire loop installation, compatible IEC 60118-4 certified loop	1	\$1,500–\$4,000	\$1,500–\$4,000	Must	Serves hearing aid and cochlear implant users with telecoil (T-coil) capability; ADA	Loop installation requires specialized installer; test field strength with IEC 60118-4-

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	amplifier, field strength target –3 to +3 dB re 100 mA/m, telecoil (T-coil) compatible, interference rejection filtering, signal input from DSP/audio system					Title III best practice; required under many state accessibility codes	compliance meter; post signage per ADA guidelines
Rack or AV Cabinet	Wall-mount ventilated rack enclosure, 12U minimum, lockable door with vented panels, contained cable management, integrated power strip (surge-protected), load rating ≥150 lbs	1	\$400–\$900	\$400–\$900	Must	Secure equipment housing prevents unauthorized tampering with accessibility systems	Verify wall construction (CMU vs. drywall) before specifying mount type; use seismic-rated hardware in applicable regions
Professional	Certified AV	1 project	\$3,500–\$8,000	\$3,500–\$8,000	Must	Installer must	Require CTS-

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
Installation (Labor)	integrator installation; includes mounting, cable runs, termination, configuration, commissioning, and user training; minimum one day on-site training included; as-built drawings and documentation required at project closeout					verify ADA-compliant control heights and accessible cable routing	certified technician; request proof of liability insurance and contract or license
Subtotal (Equipment Only)	\$10,270-\$27,200						
15% Contingency	\$1,540-\$4,080	For unforeseen conditions, additional cabling, or scope changes during installation					
Total Estimated Range (Equipment + Labor)	\$15,310-\$39,280	Includes equipment, installation labor, and contingency; excludes furniture, construction, and IT infrastructure costs					

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
+ Continuity)							

Template A-2: Hybrid / Collaborative Classroom AV Bill of Materials

Scenario: Active learning or hybrid classroom, 20–40 students, designed for simultaneous in-room and remote participant engagement, video conferencing-first design, instructor mobility, and collaborative group work.

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
Interactive Flat Panel Display (Primary)	Minimum 86-inch diagonal, 4K UHD, 450 nits minimum, 20-point multi-touch, HDMI 2.0 and USB-C inputs, built-in Android OS, OPS slot for optional PC module, front-facing stereo	1	\$4,500–\$8,000	\$4,500–\$8,000	Must	Large format supports visibility from all seating positions ; integrated captioning display	ADA mount height compliance required; verify sightline clearance from all seat positions

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	speakers (2×20W minimum)						
Video Conferencing Room System	Integrated all-in-one camera/speaker/microphone bar, 4K camera (minimum 120° horizontal FOV), built-in beamforming microphone array (6-element minimum), 8W speaker output, USB or IP connectivity, compatible with Microsoft Teams, Zoom, WebEx, and Google Meet, automatic people framing AI, privacy shutter	1	\$1,500-\$4,000	\$1,500-\$4,000	Must	Auto-framing supports equitable representation of all participants; noise suppression aids participants with auditory processing difficulties	Platform certification required; confirm compatibility with institution's UC platform
PTZ Camera	Minimum 12x	1	\$1,500-\$4,000	\$1,500-\$4,000	Should	Enables visual	Position to

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	optical zoom, pan $\pm 170^\circ$, tilt -30° to $+90^\circ$, auto-framing AI capability, USB 3.0 and IP (RTSP/NDI) output, PoE+ powered option, preset positions programmable via control system, RS-232/IP control					focus on instructor and whiteboard for remote learners; improves equity of remote experience	capture whiteboard, instructor station, and audience view; coordinate with lighting
Ceiling Microphone Array	Beamforming ceiling microphone, minimum 10-foot effective coverage radius, acoustic echo cancellation (AEC) built-in, noise suppression, USB and Dante/A	1-2 (room size dependent)	\$1,200-\$3,500 each	\$1,200-\$7,000	Must	360° pickup eliminates "must sit near the mic" barrier; critical for equitable hearing for remote participants	One unit covers approximately 15 ft x 15 ft area; larger rooms require 2 units; coordinate with HVAC locations

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	oIP output, directional steering (automatic or programmable), mute indicator (visual LED), PoE+ powered						
Confidence Monitor / Second Display	Minimum 55-inch, 4K UHD, IPS panel, HDMI 2.0 input, wall or cart mount, used for instructor confidence view (remote gallery, presentation preview, or notes)	1	\$600–\$1,500	\$600–\$1,500	Should	Allows instructor to maintain eye contact with in-room students while monitoring remote participants	Position at instructor station level; can also serve as secondary content display for students
Wireless Microphone System (2-Channel)	2-channel UHF or DECT wireless system, one handheld transmitter with cardioid	1 system	\$1,200–\$3,000	\$1,200–\$3,000	Must	Handheld enables audience Q&A participation; lavalier supports instructor mobility;	Coordinate RF frequencies with all wireless systems in building; use frequency

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	capsule, one bodypack with lavalier capsule, dual-channel receiver with diversity, rechargeable batteries , charging dock, XLR/USB output					both feed assistive listening and hearing loop systems	coordination software
Digital Signal Processor (Advanced)	Minimum 8x8 I/O DSP, built-in AEC (multi-channel) , noise gate, dynamic EQ, compressor/limiter, Dante or AES67 networking, RS-232 and IP control, compatible with major UC platforms, VoIP/SIP interface	1	\$1,500-\$4,000	\$1,500-\$4,000	Must	Central hub for all audio accessibility: hearing loop drive, ALS feed, captioning audio output	Program by certified DSP engineer ; document all routing and processing in as-built DSP file
Managed	8-port	1	\$300-	\$300-	Must	Powers	Configur

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
Network Switch	minimum, PoE+ (30W per port, 60W total budget minimum 240W), 1Gbps copper, 1× SFP uplink, Layer 2 managed (VLAN, QoS, IGMP snooping), rack-mountable, web and CLI management, AV VLAN capable		\$800	\$800		PoE assistive devices; VLAN segmentation protects AV accessibility systems from network congestion	AV VLAN with IT before installation; document VLAN ID and QoS markings in as-built
Room Scheduling Panel	7-inch color touchscreen, IP-based (PoE+ powered), integration with campus calendar (Exchange, Google Workspace, EMS), LED room status indicator	1	\$500–\$1,200	\$500–\$1,200	Should	Must include accessible LED status visible to users with low vision; mount at ADA-compliant height	Coordinate integration with campus room booking system prior to procurement

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	(visible from corridor), gang box or mullion mount, web browser UI for management						
Digital Signage Media Player	4K playback capability, network-connected (Ethernet and Wi-Fi), CMS-compatible (open API), auto-on/off scheduling, HDMI 2.0 output, fanless or quiet operation, remote management capability	1	\$300–\$700	\$300–\$700	Nice	Supports pre-class accessibility information display (e.g., captioning availability, ALS receiver pickup); WCAG-compliant content required	Requires CMS subscription; coordinate with institution's existing signage infrastructure
Noise Masking System	Sound masking emitter (in-ceiling or above-ceiling),	1 zone	\$800–\$2,500	\$800–\$2,500	Nice	Reduces distraction for students with ADHD or auditory	Only applicable if adjacent to open learning common

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	pink or speech-spectrum noise output, zone controllable, target output level 45–50 dBA at desk level, compatible with adjacent corridor and open-area noise control					sensitivity; improves speech intelligibility within the defined zone	s or high-traffic areas; tune by acoustic technician
Ceiling Speaker System (Extended)	4-speaker distributed ceiling audio configuration, 70V/100 V, minimum 8-inch drivers, back-can enclosures, matched impedance taps, coverage pattern per ANSI S12.60 design	1 system (4 spkrs)	\$800–\$1,800	\$800–\$1,800	Must	Consistent SPL across all seating positions for equitable audio access	Commission with SPL measurement at all seating rows; document in acceptance test report
AV	Centraliz	1	\$1,500–	\$1,500–	Should	Simplify	Program

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
Control System (Full)	ed control processor (IP-based), event-driven programming, RS-232, IR, relay, and IP device control, cloud management portal, third-party UC platform integration, monitoring and alerting capability		\$4,000	\$4,000		s AV operation for educators; single-button start reduces technology barrier; accessibility-mode presets programmable	source code must be provided to institution at project closeout
CAT6A Structured Cabling	As Template A-1; increased quantity for additional networked devices	8-10 drops	\$150-\$350 per drop	\$1,200-\$3,500	Must	Supports all networked accessibility devices	Plan drops for: IFP, DSP, PTZ camera, ceiling mic(s), control system, switch, scheduling panel, signage player
Assistive Listening System + Hearing	As Template A-1; FM system	1 system	\$2,500-\$6,000	\$2,500-\$6,000	Must	Required per ADA; verify IEC 60118-4	ALS receivers must be stored in accessible

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
Loop	with minimum 6 receivers plus room perimeter hearing loop					compliance of loop system	e, labeled location
Power Conditioning / UPS (Expanded)	Minimum 2200VA, pure sine wave, AVR, minimum 8 outlets, network card for monitoring, minimum 15-min runtime at 50% load	1	\$700–\$1,500	\$700–\$1,500	Must	Uninterrupted power for accessibility systems during brief outages	Calculate actual connected load before specifying UPS capacity
Rack / AV Cabinet	Wall-mount ventilated rack, minimum 18U, lockable, integrated cable management, thermal management/fan tray, 20A power strip	1	\$700–\$1,500	\$700–\$1,500	Must	Secure enclosure for accessibility system components	Allow 2U blank panels above and below heat-generating equipment
Professional	Certified AV	1 project	\$8,000–\$18,000	\$8,000–\$18,000	Must	Verify ADA	Require CTS-D or

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
Installation (Labor)	integrator; includes all mounting, cable runs, termination, DSP programming, control system programming, commissioning, acceptance testing, and minimum 2-day faculty training					compliance at closeout inspection; installer to provide accessibility feature test documentation	CTS-I certified lead technician for hybrid room complexity
Subtotal (Equipment Only)	\$22,100-\$55,800						
15% Contingency	\$3,315-\$8,370						
Total Estimated Range	\$33,415-\$82,170	Equipment + installation labor + contingency					

Template A-3: Auditorium / Large Venue AV Bill of Materials

Scenario: Large lecture hall, performance auditorium, or multi-purpose event venue serving 300–500 occupants. Designed for live events, lectures, performances, commencement ceremonies, and distance/simulcast broadcast.

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
Line Array Speaker System (L/R Main)	Minimum 4-element line array per side (L/R), rigged or ground-stacked, frequency response 60Hz–18kHz (± 3 dB), minimum 130 dB SPL peak per cluster, weatherized if outdoor use required, passive or active configuration, built-in DSP with networkable amplification	2 clusters (L/R)	\$8,000–\$30,000 per cluster	\$16,000–\$60,000	Must	Uniform SPL across all seating for equitable audio access	Hire certified rigging engineer for flown installation; per ETCP standards
Subwoof	Minimum	2–4 units	\$3,000–	\$6,000–	Should	Cardioid	Delay

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
er Array (Cardioid Configuration)	2× 18-inch subwoofers, cardioid array configuration (end-fire or arc) to control rear energy, 45Hz–100Hz bandwidth, minimum 130 dB SPL output, integrated DSP with delay and polarity control		\$8,000 each	\$32,000		pattern reduces low-frequency energy on stage (improves feedback rejection and monitor clarity)	align to main clusters during system commissioning; tune with measurement software
Digital Audio Mixing Console (FOH)	Minimum 32-channel digital mixing console, front-of-house position, built-in effects processing, motorized faders, recall scene memory, Dante/AoIP networki	1	\$8,000–\$25,000	\$8,000–\$25,000	Must	Console must have accessible fader layout; remote tablet control enables operation from accessible FOH position	Provide dedicated 20A circuit at FOH position; coordinate FOH position for sightline to stage

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	ng card, multi-track USB recording I/O, touchscreen display, remote control via tablet app						
Wireless Microphone System (8-Channel Diversity)	8-channel UHF or DECT digital wireless system, handheld and bodypack transmitters (mix as needed per application), antenna distribution system, frequency coordination software, rechargeable battery system with charging dock,	1 system	\$12,000-\$35,000	\$12,000-\$35,000	Must	Multiple simultaneous channels support panel discussions, Q&A, and equal access to the floor for all participants	Conduct frequency coordination before install; scan local RF environment; register channels if required by jurisdiction

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	diversity reception, minimum 300 ft range (open space)						
Monitor / IEM System	In-ear monitor system (8-channel stereo IEM), or minimum 4x stage floor wedge monitors (12-inch coaxial, 500W), stage monitor amplification, separate monitor mix via console aux sends or dedicated monitor desk	1 system	\$4,000–\$12,000	\$4,000–\$12,000	Must	IEM system preferred (reduces stage bleed and ambient noise, improving ALS/hearing loop signal clarity)	IEM RF frequencies must be coordinated with wireless microphone system channels
Projection System or LED Wall	Option A (Projection): Minimum 20,000 ANSI lumen laser projector, WUXGA	1 system	\$40,000–\$200,000	\$40,000–\$200,000	Must	Minimum 7:1 image contrast ratio for caption legibility; caption window must be	Engage structural engineer for LED wall or rigged projection screen; consider

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	or 4K resolution, short-throw or long-throw lens as required by throw ratio, Christie/Barco-class reliability (brand-neutral spec), rear-projection or front-projection screen 25-ft wide minimum. Option B (LED Wall): 2.5mm or finer pixel pitch indoor LED wall, minimum 18ft x 10ft, peak brightness 800 nits minimum, HDR-capable, service access from rear					visible from all seats; ADA sight-line compliance required	ambient light when specifying lumen output

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
IMAG Camera System	3-camera system: 1× main PTZ (stage center, IP/NDI output), 1× operated camera position (FOH or side balcony, SDI output), 1× wide/room shot camera (IP); all with RCP (remote control panel) capability, matching sensor size and color science; confidence preview monitors at camera positions	3 cameras	\$5,000–\$20,000 per camera	\$15,000–\$60,000	Should	IMAG (Image Magnification) provides close-up views for attendees with low vision in distant seating	Requires camera operator training; integrate with production switcher and IMAG delay for sync to PA
Video Production Switcher	Minimum 8 inputs (SDI and HDMI), 2 outputs, T-bar or auto-transition	1	\$2,000–\$12,000	\$2,000–\$12,000	Must	Enables picture-in-picture display of sign language	Locate at FOH or dedicated production table with

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	n control, downstream keyer (for graphic overlay), multiview output, USB recording output, optional NDI integration for streaming; 1RU or desktop form factor					interpreter alongside presentation content	clear sightline to stage
Assistive Listening System (FM + Hearing Loop)	FM ALS: minimum 25% of fixed seating (ADA requirement for venues with audio systems), neck loops included. Hearing Loop: full room or partial loop (perimeter or phased array as dictated	1 system	\$5,000–\$20,000	\$5,000–\$20,000	Must	ADA Title III mandatory for public assembly venues; provides equitable access for D/HH community	Post ADA-required signage at all public entrances; place receiver storage and checkout at accessible counter height

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	by room geometry), IEC 60118-4 certified amplifier and loop wire installation						
Live Streaming Encoder (Hardware)	Hardware encoder, minimum 1080p/60fps output, dual streaming capability (primary and backup destination), H.264 and H.265 encoding, direct RTMP/RTSP output, XLR audio input (2-channel minimum), HDMI/SDI video input, rackmount form factor, front-	1	\$2,000-\$8,000	\$2,000-\$8,000	Should	Live stream enables remote access for attendees with mobility disabilities who cannot attend in person; requires closed captions on stream (WCAG 2.1 1.2.4)	Configure automatic caption passthrough or integrate with CART service for live stream captions

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	panel monitoring						
Production Intercom System	2-channel wired intercom (partyline), minimum 4 beltpacks, headsets included, power supply beltpack, XLR connectivity, clear-com or equivalent party-line topology, stage manager and FOH positions minimum	1 system	\$2,000-\$6,000	\$2,000-\$6,000	Should	Enables coordinated cue calls including accessibility cue points (interpreter position, ALS distribution)	Include intercom drop at lighting board position, followspot positions, stage manager's desk
Stage Manager's Monitor	Minimum 24-inch monitor at stage manager's position, showing multiview of all camera feeds and program output,	1	\$400-\$1,000	\$400-\$1,000	Should	Enables monitoring of IMAG and captioning display for quality assurance during events	Position at stage left or right wing per production convention

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	HDMI or SDI input, front-facing speaker for program audio reference						
Multi-Track Recording System	32-channel or greater digital audio recording interface , USB or Dante connectivity to FOH console, SSD-based recording, compatible with standard DAW software , simultaneous audio+video recording capability (1080p minimum video)	1	\$3,000–\$10,000	\$3,000–\$10,000	Nice	Post-event recordings enable on-demand access for students who cannot attend in person; requires caption addition for post-production accessibility compliance	Establish archiving policy before purchase ; SSD storage plan required
Stage Box /	Minimum 16-	1	\$3,000–\$8,000	\$3,000–\$8,000	Must	Eliminates long	Digital snake

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
Digital Snake	channel stage box (8 in/4 out minimum), digital snake over CAT6/fiber to FOH, Dante/AoIP compatible, locking panel connectors (XLR female in, XLR male out), tie-line path to monitor position					analog cable runs that degrade audio quality and hearing loop signal	dramatically reduces cable infrastructure cost and weight compared to analog snake
Professional Installation (Labor)	Certified AV integrator (CTS-I preferred); full rigging, cabling, commissioning, acoustic measurement, system tuning, operator training (minimum 2 days), and as-	1 project	\$30,000–\$80,000	\$30,000–\$80,000	Must	ADA compliance inspection mandatory at closeout; hearing loop field strength tested and documented per IEC 60118-4	ETCP-certified rigger required for any flown speaker or projection systems

Item Category	Specification Description (Performance-Based)	Qty	Est. Unit Cost Range	Est. Total Range	Priority	Accessibility Feature	Notes
	built documentation; structural engineering sign-off for rigged elements						
Subtotal (Equipment Only)	\$136,400-\$476,000						
15% Contingency	\$20,460-\$71,400						
Total Estimated Range	\$186,860-\$627,400	Wide range driven by projection vs. LED wall decision and IMAG camera system scope					

Section B: Site Survey Templates

A thorough site survey is the essential first step in any AV planning or upgrade project. The templates in this section provide standardized, reproducible data collection forms that can be used by AV coordinators, instructional technology staff, and integrators to document existing conditions, identify constraints, and establish the basis for system design. Completing these surveys before developing any specifications or budget estimates will significantly improve project accuracy and reduce costly change orders during installation.

Template B-1: Classroom AV Site Survey Form

SITE SURVEY HEADER					
Date of Survey:	_____	Surveyor Name:	_____	Surveyor Title:	_____
Institution :	_____	Building:	_____	Room Number/ID:	_____
Room Primary Use:	_____	Typical Occupancy:	_____	Survey Type:	☐ New Installation ☐ Upgrade ☐ Assessment

Room Physical Data

Measurement	Value	Notes
Room Length (ft)	_____	
Room Width (ft)	_____	
Ceiling Height (ft)	_____	
Room Volume (cu ft)	_____ (L × W × H)	Used to estimate reverberation characteristics
Floor Area (sq ft)	_____ (L × W)	Used for seating capacity and speaker coverage calculations
Ceiling Type	_____	Acoustic tile / Drywall / Exposed concrete / Grid / Other: _____
Floor Type	_____	Carpet / Hard surface / Mixed / Raised floor
Wall Construction	_____	Drywall / CMU / Glass curtain / Combination: _____
Window Coverage (%)	_____	Percentage of display wall with adjacent windows
Blinds / Shades Present	☐ Yes ☐ No	Type: _____; Motorized: ☐

Measurement	Value	Notes
		Yes ☐ No
Column Obstructions	☐ Yes ☐ No	Number: ____ Location(s): _____
Raised Floor or Trench	☐ Yes ☐ No	Useful for cabling routing
ADA Pathway to Control Location	☐ Clear ☐ Obstructed	Note obstructions: _____

Lighting Conditions

Location	Measured Lux	Time of Day	Window Condition	Notes
At Display Wall (primary screen position)			☐ Shaded ☐ Open ☐ N/A	
At Student Desks (front row)				
At Student Desks (rear row)				
At Instructor Station				
Peak Ambient Lux (worst case location)				Use for minimum display brightness calculation
Lighting Control Available	☐ Manual switch ☐ Dimmer ☐ Occupancy sensor ☐ None	Zones: _____		

Acoustic Conditions

Measurement	Value	Standard / Target	Pass / Fail
Ambient Noise Level (unoccupied, all HVAC on, dBA)		≤35 dBA (ANSI S12.60)	☐ Pass ☐ Fail

Measurement	Value	Standard / Target	Pass / Fail
Dominant Noise Source		Identify: HVAC / Exterior traffic / Adjacent room / Other	
HVAC Noise Impact (subjective)		None / Mild / Significant	
Estimated RT60 (clap test, seconds)		≤0.6 seconds (ANSI S12.60 for classroom)	☐ Pass ☐ Fail
Adjacent Space Noise Intrusion		None / Mild / Significant	
Wall/Door Sound Isolation		Identify gaps at doors, penetrations	
Surface Reflections (hard parallel walls)	☐ Present ☐ Not present	Can cause flutter echo; mitigation recommended	

Existing AV Equipment Inventory

Item Type	Location	Make/ Model (if visible)	Condition (1-5)	Age (approx.)	Notes / Retain or Replace
Display 1 (primary)					
Display 2 (secondary)					
Projector					Lamp hours if accessible: _____
Audio System / Speakers					
Microphone System					
Control System / Touchpanel					
Switcher / Matrix					
Cabling (visible /					Type: HDMI / VGA / CAT /

Item Type	Location	Make/ Model (if visible)	Condition (1-5)	Age (approx.)	Notes / Retain or Replace
accessible)					Other: ____
Assistive Listening System					Number of receivers: _____
Hearing Loop					Loop wire visible: [] Y [] N
Document Camera					
Other (specify)					

Power and Network Infrastructure

Infrastructure Item	Quantity	Location(s)	Capacity / Speed	Notes
Standard 120V Outlets (15A)			15A	Note NEMA configuration
20A Dedicated Circuit (AV use)			20A	Panel location: ____; Breaker ID: ____
Network (CAT) Drops			Tested speed: ___ Mbps	CAT5e / CAT6 / CAT6A; patch panel location: _____
Wi-Fi Coverage (AV-band)			Signal: ____ dBm; Band: 2.4 / 5 GHz	Confirm 5GHz coverage at instructor and student areas
Fiber Runs (if present)			Type: SM / MM; Connectors: _____	
Conduit (empty or occupied)			Size: ____ inches	Fill percentage estimate: _____
Ground Bus / Building Ground	[] Present [] Not located			Critical for ground loop prevention

Accessibility Features Checklist

- Assistive listening system (FM or infrared) present and functional
- Hearing loop (induction loop) installed; T-coil signage posted
- Captioning display capability available (display or dedicated caption monitor)
- AV controls mounted at ADA-compliant height (≤ 48 " AFF forward reach, ≤ 54 " side reach)
- Clear accessible pathway (36" min.) to all AV controls and instructor station
- Braille signage present at room entrance (room number)
- Accessible seating locations designated (D/HH, mobility, sightline)
- Audio description output available (or input provided for AD system)
- Large-print or digital room guide available
- Lighting controls accessible from ADA-compliant location
- Emergency notification visual alert (strobe) visible from all seating positions

Survey Narrative and Recommendations

Field	Entry
Current State Summary	
Primary Issues Identified	
Recommended AV Configuration	
Estimated Budget Range	\$ _____ to \$ _____
Priority Level for Upgrade	☐ Immediate ☐ Short-term (within 1 year) ☐ Long-term (2-3 years)
Surveyor Signature	Date: _____

Template B-2: Multi-Room Campus Survey

Summary Matrix

Use this matrix to consolidate individual room survey data from Template B-1 into a campus-wide prioritization tool. Enter each surveyed room as a row. Use this matrix for budget prioritization, grant applications, and administrative reporting.

Room ID	Room Size (sq ft)	Ceiling Ht (ft)	Display Type	Audio Quality (1-5)	Control System (Y/N)	Network Speed (Mbps)	Assistive Listening (Y/N)	ADA Compliant (Y/N/Partial)	Upgrade Priority (H/M/L)	Est. Budget Range	Notes
____ ____											
____ ____											
____ ____											
____ ____											
____ ____											
____ ____											
____ ____											
Campus Totals / Averages											

Audio Quality Rating Scale (1-5)

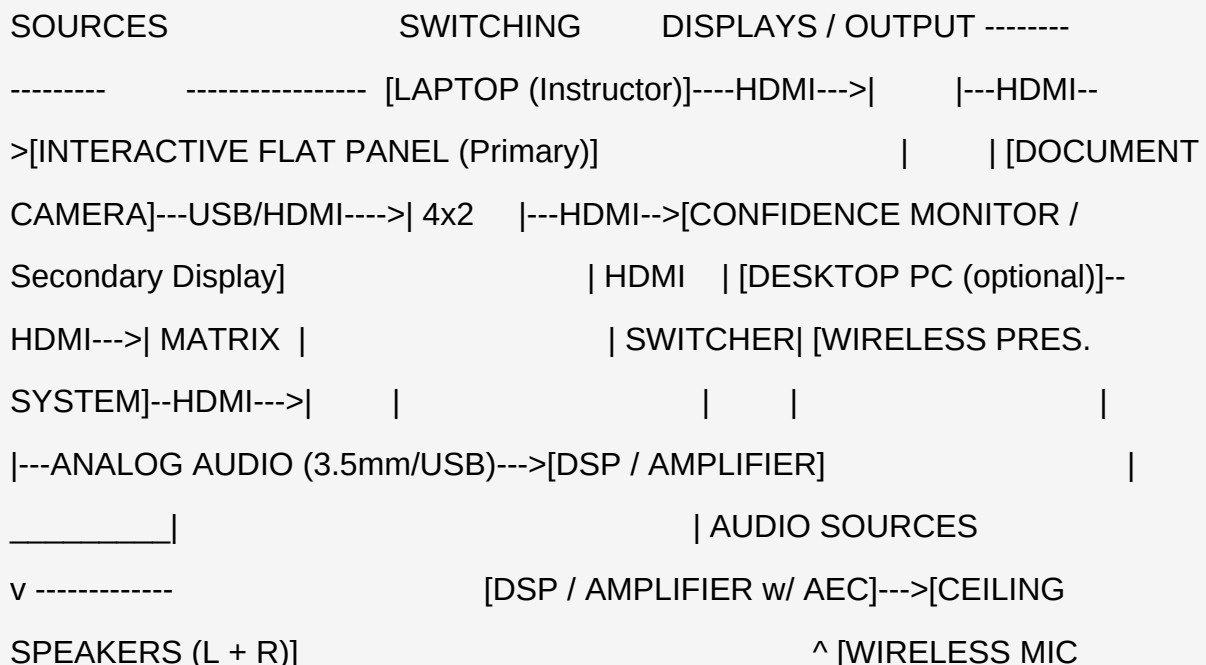
1 - Poor: Significant ambient noise, HVAC intrusion, poor intelligibility; RT60 >1.0 sec.
 2 - Below Standard: Noticeable acoustic issues; RT60 0.8–1.0 sec.

3 - Acceptable: Marginally compliant; RT60 0.6–0.8 sec. **4 - Good:** Meets ANSI S12.60; RT60 <0.6 sec. **5 - Excellent:** Exceeds standard; treated surfaces, measured STI >0.75.

Section C: Signal Flow Diagram Templates and Descriptions

Signal flow diagrams are fundamental tools for AV design, documentation, troubleshooting, and training. The diagrams in this section use a text/ASCII representation suitable for inclusion in this document and for instructor use on whiteboards. For production drawings, recreate these diagrams in draw.io, Visio, or similar diagramming tools using standard AV symbol libraries. Each diagram is followed by a component legend and signal type reference table.

C-1: Basic Classroom Signal Flow



SYSTEM]---RF Receiver---XLR----->| [HEARING LOOP AMPLIFIER INPUT]

<-----{AUX data-id="1060" OUTPUT from DSP) CONTROL ----- [AV CONTROL TOUCHPANEL (7")]---IP/CAT6--->[AV CONTROL PROCESSOR]

|--- RS-232/IP -->[MATRIX SWITCHER] |--- RS-232/IP -->[DSP/AMPLIFIER] |--- CEC/HDMI ---

>[FLAT PANEL DISPLAY] |--- IR/IP -----

>[DOCUMENT CAMERA] |--- IP -----

>[WIRELESS PRES. SYSTEM] ACCESSIBILITY PATH ----- [DSP AUX OUTPUT]---LINE LEVEL--->[FM ALS TRANSMITTER]---RF--->[FM RECEIVERS (worn by D/HH users)] [DSP AUX OUTPUT]---LINE LEVEL--->[HEARING LOOP AMPLIFIER]---CURRENT--->[LOOP WIRE (room perimeter)]

| [Telecoil in hearing aid/CI picks up field]

C-1 Component Legend

Symbol / Label	Component Type	Function	Signal Type In	Signal Type Out
LAPTOP	Source Device	Instructor content origination point	N/A (origin)	HDMI (video+audio), USB-C
DOCUMENT CAMERA	Source Device	Displays physical documents and 3D objects on screen	N/A (optical capture)	USB 3.0, HDMI
DESKTOP PC	Source Device	Permanent installed computing source	N/A (origin)	HDMI, DisplayPort
WIRELESS PRES. SYSTEM	Source / Wireless Gateway	Allows BYOD devices to share content wirelessly	Wi-Fi (2.4/5 GHz) from BYOD devices	HDMI (decoded output to switcher)
4x2 HDMI	Active Signal	Routes any of 4	4x HDMI in	2x HDMI out;

Symbol / Label	Component Type	Function	Signal Type In	Signal Type Out
MATRIX SWITCHER	Routing Device	video sources to any of 2 display outputs simultaneously		analog audio breakout
INTERACTIVE FLAT PANEL	Display (Active)	Primary classroom display with touch interactivity	HDMI, USB-C	HDMI loop-out (to secondary display)
CONFIDENCE MONITOR	Display (Active)	Secondary display showing same or alternate source to instructor or students	HDMI	N/A
DSP / AMPLIFIER	Active Signal Processing	Processes audio: EQ, AEC, noise suppression, routing, amplification	XLR (mic), analog (line), USB, AoIP	70V speaker output; AUX out to ALS/loop
WIRELESS MIC SYSTEM	Active Transducer / RF System	Captures instructor voice wirelessly; transmits to DSP	RF (wireless) from bodypack transmitter	XLR or USB audio to DSP
CEILING SPEAKERS	Passive Transducer	Converts amplified audio signal to sound pressure; distributes evenly in room	70V amplified audio from DSP	N/A (acoustic)
AV CONTROL TOUCHPANEL	Control Interface (Active)	User interface for faculty to operate AV system	Touch input from user	IP commands to control processor
AV CONTROL PROCESSOR	Active Control Engine	Executes programmed control logic; sends commands to all AV devices	IP from touchpanel	RS-232, IP, IR, relay to all devices
FM ALS TRANSMITTER	Active Accessibility Device	Transmits audio signal as FM radio to ALS receivers worn by D/HH users	Line-level audio from DSP aux	RF (72-76 MHz FM)

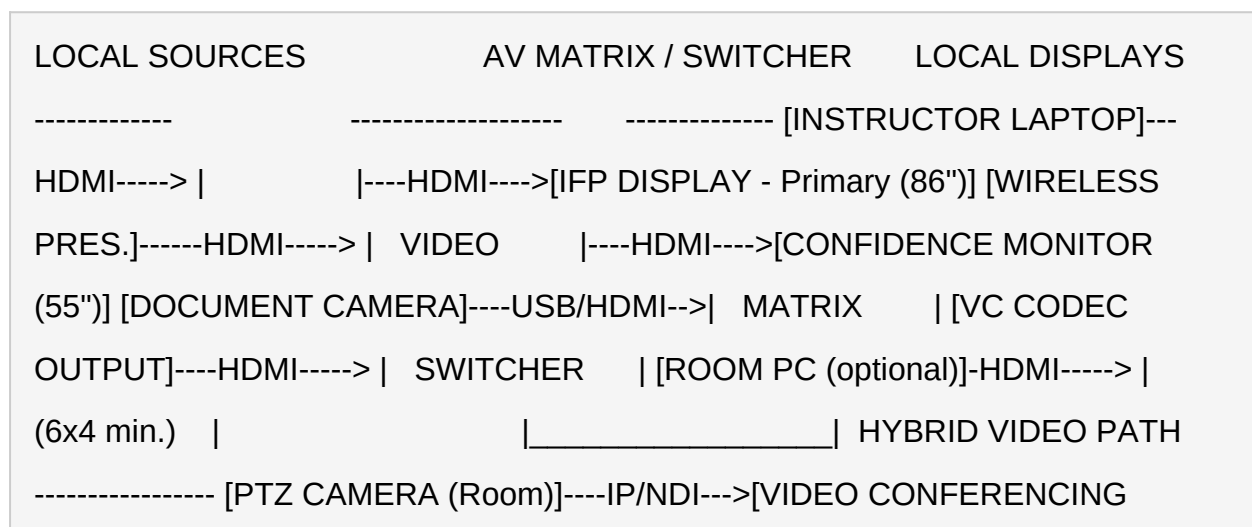
Symbol / Label	Component Type	Function	Signal Type In	Signal Type Out
HEARING LOOP AMPLIFIER	Active Accessibility Device	Drives current through loop wire to create magnetic field compatible with T-coil hearing aids	Line-level audio from DSP aux	Current loop (electromagnetic field)

C-1 Signal Type Reference

Signal Type	Typical Connector	Active or Passive	Max Practical Distance	Notes
HDMI 2.0	HDMI Type A	Passive (standard) / Active (extender)	3m passive; extended with active or HDBaseT	Carries digital video + audio + HDCP; EDID management required
HDBaseT	RJ-45 (CAT6A)	Active (requires powered transmitter/receiver)	100m (328 ft)	Extends HDMI over structured cabling; PoE option for remote device power
USB 3.0/3.1	USB Type A/B/C	Active	5m; extended with active cables or hubs	Carries data, video, audio, and power (USB-C); universal device compatibility
XLR (Balanced)	XLR 3-pin	Passive	300m+ for mic-level; noise-immune	Professional audio standard; balanced differential signal rejects interference
TRS (Balanced Line)	6.35mm or 3.5mm TRS	Passive	30m (line-level balanced)	Common for consumer/prosumer interconnection; more susceptible to noise than XLR
70V Audio	Speaker wire (terminals)	Requires 70V amplifier	300m+	Constant-voltage distribution

				system; each speaker has a transformer tap
RS-232	DB-9 or RJ-45 serial	Active	15m standard; extended with converter	Bidirectional serial control; widely used for display and switcher control
IP (Ethernet)	RJ-45 / CAT6A	Active (requires network infrastructure)	100m per segment	Primary modern AV control and networked media transport protocol
IR (Infrared)	3.5mm mini-jack (emitter)	Active (modulated IR signal)	Line-of-sight; emitter wired to device	Legacy control standard; limited feedback; being replaced by IP control in modern installations
RF (Radio Frequency)	BNC / SMA antenna	Active	30-100m depending on frequency and power	Used by wireless microphone and FM ALS systems; requires frequency coordination

C-2: Hybrid Classroom Signal Flow



CODEC/SYSTEM] [VC BAR CAMERA]----USB/IP----->| (Software or Hardware)

|---IP/SIP/H.323--->[CLOUD PLATFORM] | |

| |

<---Remote data-id="1223" participant video---[CLOUD PLATFORM]

<---[REMOTE PARTICIPANTS] | |---HDMI

(remote gallery output)--->[AV MATRIX (remote feed input)] AUDIO PATH

(HYBRID) ----- [CEILING MIC ARRAY (beamforming)]-----USB/Dante--

>[DSP (w/ AEC)]-->[VC CODEC (mic input)] [WIRELESS MIC SYSTEM (2-ch)]---

XLR/Dante----->| | [VC BAR MICROPHONE]---USB-----

>| DSP PROCESSOR | | | + AEC/AGC | |

| | | [VC CODEC

SPEAKER OUTPUT]---XLR----->[DSP ROUTING] |

| v [CEILING SPEAKERS (4x)] [VC CODEC]--IP--

>[REMOTE AUDIO] |

[FM ALS TRANSMITTER]

<--AUX data-id="1224" OUT [HEARING LOOP AMP]

<---AUX OUT CONTROL AND NETWORK ----- [AV TOUCHPANEL

(7")]---IP/PoE--->[AV CONTROL PROCESSOR] |--RS-

232/IP-->[MATRIX SWITCHER] |--Dante/IP--->[DSP

PROCESSOR] |--IP----->[PTZ CAMERA (presets)]

|--IP----->[VC CODEC/SYSTEM] |--IP----->[ROOM

SCHEDULING PANEL] |--IP----->[DIGITAL SIGNAGE

PLAYER] [MANAGED NETWORK SWITCH (PoE+)]---provides power + data to:

| - CEILING MIC ARRAY(s) | - ROOM SCHEDULING PANEL | - AV CONTROL

PROCESSOR | - VC CODEC (if IP-connected) | - PTZ CAMERA | - DSP (if

Dante/IP) VLAN CONFIGURATION (required): VLAN 10: AV Devices (DSP,

Switcher, Control, PTZ, VC Codec) VLAN 20: Wireless Presentation / BYOD

VLAN 1 (data): Campus standard IT network QoS marking: AV traffic = DSCP EF (46); VC traffic = DSCP AF41

C-3: Auditorium / Production Signal Flow

STAGE SOURCES	STAGE BOX / DIGITAL SNAKE	FOH MIX
POSITION -----	-----	----- [WIRED
PODIUM MIC (XLR)]----->	[WIRELESS MIC RX (8-ch)]--XLR-->	
STAGE BOX	---DANTE/AES67-IP--->[FOH MIX CONSOLE (32ch)]	
[INSTRUMENT DI BOXES]---TRS/XLR>	(16in/8out min.)	
[PLAYBACK SYSTEM]---XLR----->	Digital Snake	
[HEARING LOOP FEED]		
<----XLR----- data-id="1227" over CAT6/Fiber		FOH PROCESSING PATH:
[STAGE MONITOR FEED]		
<---XLR-----	[INSERT: Comp/Limit/Gate]	
-----	[OUTPUT EQ + DSP]	
	MAIN PA OUTPUT PATH	v
-----	[AMPLIFIER NETWORK (DSP + Amp)]	
<-+ [FOH CONSOLE MAIN L/R OUT]---AES67/Dante OR XLR----->[DSP +		
AMPLIFICATION]		[LINE
ARRAY - LEFT CLUSTER]		
<--- data-id="1228"	[LINE ARRAY - RIGHT CLUSTER]	
<--	[SUBWOOFER ARRAY (cardioid)]	
<--	[FRONT FILL SPEAKERS (stage)]	

```

<-| [DELAY SPEAKERS (rear zones)]

<-| MONITOR / IEM PATH ----- [FOH/Monitor CONSOLE AUX SENDS]---
Dante--->[IEM TRANSMITTER RACK (8-ch)]---RF--->[IEM BELTPACKS
(performers)] |----->[STAGE MONITOR AMPLIFIERS]-----
>[FLOOR WEDGE MONITORS] |----->[STAGE MANAGER
MONITOR SPEAKER] IMAG VIDEO PATH ----- [PTZ CAMERA - Stage
Center]---IP/NDI--->[VIDEO PRODUCTION SWITCHER]---HDMI/SDI--->[MAIN
PROJECTION/LED WALL] [OPERATED CAMERA - FOH]-----SDI----->| (8-input
minimum) | [WIDE/ROOM CAMERA]-----IP/NDI--->| |---
SDI--->[RECORD / STREAM OUTPUT] [PRESENTATION PC (slides)]--HDMI----->|
|---SDI--->[STAGE MANAGER MONITOR] [BACKUP LAPTOP]-----HDMI-----
>|_____| STREAMING AND RECORDING PATH
----- [VIDEO SWITCHER PROGRAM OUT]---SDI--->[HARDWARE
ENCODER]---IP--->[STREAMING CDN (primary)] |
|---IP--->[STREAMING CDN (backup)] | |---
USB--->[LOCAL RECORDING (SSD)] [FOH CONSOLE DANTE MULTITRACK]---
IP--->[MULTI-TRACK RECORDER (32-ch)]---USB--->[DAW WORKSTATION]
ACCESSIBILITY PATH ----- [FOH CONSOLE AUX OUT (ALS)]---XLR---
>[FM ALS TRANSMITTER]---RF--->[FM RECEIVERS (25%+ seating)] [FOH
CONSOLE AUX OUT (LOOP)]--XLR--->[LOOP AMPLIFIER]---CURRENT---
>[PERIMETER LOOP WIRE (full venue)] [CAPTIONING STATION INPUT]---USB---
>[CART PROVIDER WORKSTATION]---IP--->[CAPTION DISPLAY ON SCREENS]
[SIGN LANGUAGE INTERPRETER POSITION]---illuminated, [VIDEO SWITCHER
PIP KEY]--->[IMAG/Stream] PRODUCTION INTERCOM ----- [STAGE
MANAGER BELTPACK]---XLR Partyline--->[INTERCOM POWER SUPPLY]---
>[FOH OPERATOR BELTPACK] ---
>[LIGHTING BOARD BELTPACK] ---
>[FOLLOWSPOT POSITIONS] ---

```

Section D: Comprehensive Glossary (150+ Terms)

This glossary provides definitions for 150+ terms used throughout the *AV Fundamentals for Educators* textbook and companion documents. Terms are organized alphabetically. The Chapter Reference column indicates the primary chapter(s) where each term appears. Related Terms support cross-referencing and vocabulary building. This glossary is suitable for use as a student reference handout and as an instructor resource for pre-teaching vocabulary.

Usage Note

Definitions in this glossary are written for an educational audience at an approximate Flesch-Kincaid Grade Level 10 target. Technical accuracy has been prioritized while maintaining accessibility for readers new to the AV field. Where industry standards are referenced (ANSI, IEC, ISO), the citation reflects the standard as of 2025–2026.

A – B

Term	Definition	Chapter Reference	Related Terms
ADA (Americans with Disabilities Act)	A U.S. federal civil rights law enacted in 1990 that prohibits discrimination based on disability in public accommodations, employment, transportation, and communications. In	Ch. 11	Section 504, Section 508, assistive listening system, hearing loop, captioning

Term	Definition	Chapter Reference	Related Terms
	educational AV, the ADA establishes requirements for physical accessibility of controls, assistive listening systems in assembly areas, captioning access, and accessible pathways to AV equipment. ADA Title II covers public entities including public schools and universities; Title III covers private businesses and private schools.		
AEC (Acoustic Echo Cancellation)	A digital signal processing technique that removes the echo caused when a microphone picks up the audio signal being played through nearby loudspeakers. AEC is essential in video conferencing rooms where the far-end caller's voice, played through the room speakers, would otherwise be captured by the microphone and returned to the far end as a distracting echo. Modern AEC algorithms use reference signals from the loudspeaker output to predict and subtract the echo component from the microphone input.	Ch. 2, Ch. 7	DSP, gain structure, video conferencing, beamforming microphone
AoIP (Audio over IP)	The transmission of uncompressed or lightly compressed audio signals over	Ch. 2, Ch. 5	Dante, AVB, AES67, network switch, VLAN, QoS

Term	Definition	Chapter Reference	Related Terms
	standard IP (Ethernet) networks. AoIP protocols such as Dante, AES67, and AVB replace traditional point-to-point analog or digital audio cabling with network-based audio routing. A single CAT6 cable can carry hundreds of channels of AoIP audio simultaneously, dramatically simplifying installation in large or complex AV systems. AoIP networks require proper configuration including QoS, IGMP snooping, and network latency management.		
Amplifier	An electronic device that increases the power level of an audio signal to drive loudspeakers at usable volume levels. In educational AV systems, amplifiers may be integrated into DSPs, interactive flat panels, or installed as standalone rack-mounted units. Amplifiers are characterized by their output power (in watts), impedance compatibility, and whether they operate in constant-voltage (70V/100V) or low-impedance (4–8 ohm) modes. Constant-voltage amplifiers are	Ch. 2	DSP, 70V audio, gain structure, speaker placement

Term	Definition	Chapter Reference	Related Terms
	standard in distributed audio systems such as school hallways and classrooms.		
Aspect Ratio	The proportional relationship between the width and height of a video image or display screen, expressed as width:height (e.g., 16:9 or 4:3). The 16:9 aspect ratio is the current standard for HD and 4K content and is the native format of virtually all modern flat panel displays and projectors. Content created for one aspect ratio that is displayed on a screen with a different ratio may appear with black bars (letterboxing or pillarboxing) or may be stretched, affecting legibility and accessibility of on-screen text.	Ch. 3	resolution, display, projector, throw ratio
Assistive Listening System (ALS)	A technology system designed to help people with hearing loss receive a clearer audio signal by bypassing background noise and distance. The three primary ALS technologies are FM (radio frequency), infrared (IR optical), and induction loop (magnetic). The ADA requires assistive listening systems in assembly areas that have audio amplification	Ch. 2, Ch. 11	hearing loop, telecoil, FM system, ADA, DSP

Term	Definition	Chapter Reference	Related Terms
	systems, with a minimum number of receivers based on seating capacity. ALS systems receive a dedicated audio feed from the room's DSP or mixing console and transmit it directly to a listener's receiver or hearing aid.		
Audio Description	A narrated track added to video content that verbally describes visual elements—such as actions, settings, expressions, and on-screen text—for the benefit of viewers who are blind or have low vision. In educational AV, audio description enables access to instructional video content for students with visual disabilities. WCAG 2.1 Success Criterion 1.2.5 requires audio description for all pre-recorded synchronized media. Audio description is distinct from closed captioning, which serves people with hearing disabilities.	Ch. 3, Ch. 11	WCAG 2.1, UDL, captioning, accessibility
AVB (Audio Video Bridging)	A set of IEEE 802.1 Ethernet standards that provide time-synchronized, low-latency audio and video transport over standard Ethernet networks. AVB (also referred to as IEEE 802.1BA)	Ch. 5	AoIP, Dante, AES67, managed network switch, latency

Term	Definition	Chapter Reference	Related Terms
	guarantees delivery timing for AV streams, making it suitable for professional applications where tight synchronization between audio and video is critical. AVB requires AVB-compatible network switches throughout the signal path, distinguishing it from Dante, which can operate over standard managed switches.		
Balanced Audio	An audio interconnection method that uses three conductors—positive, negative (inverted), and ground—to transmit an audio signal in a way that rejects externally induced noise (common-mode rejection). The destination device subtracts the inverted signal from the positive signal, doubling the desired audio level while canceling any noise induced equally on both conductors. Balanced connections use XLR or TRS connectors and are the standard for professional audio equipment, microphone cables, and long cable runs in installed AV systems.	Ch. 2, Ch. 4	XLR connector, TRS connector, unbalanced audio, ground loop
Bandwidth	In networking contexts, bandwidth	Ch. 5	network switch, QoS, frequency

Term	Definition	Chapter Reference	Related Terms
	refers to the maximum data transfer capacity of a network connection, typically measured in megabits per second (Mbps) or gigabits per second (Gbps). In audio contexts, bandwidth describes the range of frequencies a system can reproduce, measured in hertz (Hz). In both contexts, sufficient bandwidth is critical for high-quality AV system performance: video conferencing requires consistent minimum bandwidth (typically 1–10 Mbps per session), and audio systems with wide frequency bandwidth reproduce sound more accurately and naturally.		response, video conferencing
Beamforming Microphone	A ceiling-mounted or table-top microphone array that uses digital signal processing to steer acoustic pickup patterns (beams) toward active speakers in a room, while simultaneously suppressing noise from other directions. Unlike fixed-pattern microphones, beamforming microphones adapt in real time, automatically following a speaker	Ch. 2, Ch. 7	polar pattern, DSP, AEC, hybrid classroom, ceiling microphone

Term	Definition	Chapter Reference	Related Terms
	as they move. They are particularly effective in active learning classrooms and hybrid environments because they eliminate the need for instructors to stay in a fixed location near a traditional microphone.		
BOM (Bill of Materials)	A structured document listing all equipment, materials, hardware, and labor items required to complete an AV installation project. A well-written BOM is vendor-neutral, specification-based (describing required performance rather than specific brands), and includes quantities, estimated costs, and priority designations. BOMs serve as the foundation for procurement solicitations (RFPs, bids) and as the primary reference document during installation and commissioning to verify that all specified components have been installed and configured.	Ch. 8	site survey, procurement checklist, RFP, signal flow
Boundary Microphone	A microphone with a flat base designed to be placed on a surface (table, floor, or wall), where it uses the boundary acoustic effect to	Ch. 2	microphone, polar pattern, omnidirectional microphone, phase

Term	Definition	Chapter Reference	Related Terms
	produce a coherent, phase-consistent pickup of sound from above the surface. Boundary microphones are commonly used on conference room tables, at the edge of stages, and at podiums. Because they sit on a boundary surface, they eliminate the acoustic interference pattern (comb filtering) that occurs when a conventional microphone is placed near a reflective surface.		
Braille Signage	Room identification signage that includes both raised tactile letters and Grade 2 Braille, required by the ADA for permanent rooms and spaces. ADA Standards for Accessible Design specify the location (mounted on the latch side of the door, 60 inches AFF to the centerline), character height, font type (sans-serif raised), and Braille cell specifications. In the context of educational AV, ALS receiver pickup locations, hearing loop zones, and room numbering must comply with ADA Braille signage requirements.	Ch. 11	ADA, accessibility, assistive listening system, hearing loop

C - D

Term	Definition	Chapter Reference	Related Terms
CART (Communication Access Realtime Translation)	A real-time stenographic captioning service in which a trained human stenographer transcribes speech to text at speeds of 225+ words per minute, with accuracy rates exceeding 98%. CART output is displayed on a screen for the benefit of Deaf and hard-of-hearing participants. CART is considered the gold standard for live captioning in educational settings and is required when automatic speech recognition (ASR) quality is insufficient for understanding. Remote CART service, delivered via IP, is increasingly common in educational institutions.	Ch. 7, Ch. 11	captioning, ASR, WCAG 2.1, accessibility, Section 504
CAT6 / CAT6A	Category 6 and Category 6A are standardized specifications for twisted-pair copper cabling used in Ethernet networks and increasingly as the transport medium for AV signal extenders (HDBaseT, AoIP). CAT6 supports 10GbE to 55 meters and is the minimum recommended for	Ch. 4, Ch. 5	HDBaseT, PoE, structured cabling, signal extender, plenum-rated cable

Term	Definition	Chapter Reference	Related Terms
	new AV installations. CAT6A (Augmented Category 6) supports 10GbE to 100 meters and is the preferred standard for AV infrastructure due to its superior alien crosstalk performance and suitability for HDBaseT. Plenum-rated (CMP) versions are required in air-handling spaces.		
Captioning (Closed, Open, Real-Time)	The display of spoken audio content as on-screen text, providing access for Deaf and hard-of-hearing viewers. Closed captions are hidden by default and activated by the viewer; open captions are permanently visible on the video frame. Real-time captioning (including CART and ASR-based captions) is generated during live events. WCAG 2.1 requires closed captions for all pre-recorded synchronized media (1.2.2) and real-time captions for all live audio content that is broadcast (1.2.4). Caption quality standards include accuracy, synchronization, completeness, and proper placement.	Ch. 3, Ch. 7, Ch. 11	CART, WCAG 2.1, ADA, Section 508, audio description
Cardioid	A heart-shaped microphone polar (pickup) pattern that is most sensitive to	Ch. 2	polar pattern, omnidirectional microphone, gain structure, feedback

Term	Definition	Chapter Reference	Related Terms
	sound arriving from the front (on-axis), less sensitive to sound from the sides, and significantly rejects sound from the rear (typically 25 dB of rear rejection). Cardioid microphones are the most common pattern in educational AV, used in wireless lavalier and handheld systems. They provide better gain-before-feedback than omnidirectional patterns in reinforced audio systems and help isolate the instructor's voice from room noise and speaker output.		
Codec (Audio)	Short for Coder-Decoder, an audio codec is a hardware or software algorithm that encodes (compresses) and decodes (decompresses) audio data. In AV systems, audio codecs determine how audio is compressed for storage or transmission. Common audio codecs include AAC (used in streaming), Opus (used in WebRTC), and PCM (uncompressed, used in professional AV). Codec selection involves tradeoffs between audio	Ch. 2, Ch. 7	codec (video), latency, bandwidth, video conferencing, WebRTC

Term	Definition	Chapter Reference	Related Terms
	quality, latency, and bandwidth consumption. In video conferencing, the audio codec directly affects call quality and intelligibility.		
Codec (Video)	A compression algorithm used to encode video data for storage or transmission and decode it for playback. H.264 (AVC) is the dominant standard for video conferencing and streaming at 1080p; H.265 (HEVC) provides approximately 50% better compression efficiency at equivalent quality, enabling 4K streaming at practical bandwidths. Video codecs determine image quality, encoding latency, and bandwidth requirements in educational AV systems including video conferencing, lecture capture, and digital signage.	Ch. 3, Ch. 7	H.264, H.265, bandwidth, video conferencing, streaming
Color Temperature	A characteristic of light and display output measured in degrees Kelvin (K) that describes the appearance of white light on a warm-to-cool spectrum. Lower color temperatures (2700K-3000K) appear warm and yellowish; higher	Ch. 3	display calibration, lumen, contrast ratio, flat panel display

Term	Definition	Chapter Reference	Related Terms
	temperatures (5000K–6500K) appear cool and blue-white. Display calibration in educational AV targets D65 (6500K) for accurate white reproduction aligned with video production standards. Projectors and flat panels that drift in color temperature over time can compromise caption legibility and color-coded instructional materials.		
Condenser Microphone	A microphone that operates using a capacitor (condenser) element consisting of a thin diaphragm positioned near a rigid backplate. Sound pressure causes the diaphragm to vibrate, changing the capacitance and generating an electrical signal. Condenser microphones require external power (phantom power at 48V DC, or battery) and are known for their high sensitivity, extended high-frequency response, and low self-noise, making them the preferred choice for speech pickup in educational environments including ceiling array microphones and studio	Ch. 2	phantom power, dynamic microphone, boundary microphone, sensitivity

Term	Definition	Chapter Reference	Related Terms
	recording.		
Contrast Ratio	A measure of a display's ability to produce both bright highlights and deep dark areas, expressed as the ratio of luminance of the brightest white to the darkest black the display can produce (e.g., 1000:1). Higher contrast ratios improve legibility of text, video detail in dark areas, and caption visibility. In real-world classroom environments, ambient light dramatically affects perceived contrast (ANSI contrast ratio vs. native contrast). Minimum 7:1 ANSI contrast ratio is recommended for educational AV display applications with ambient lighting.	Ch. 3	display, lumen, ANSI, color temperature, captioning
Control Processor	The central computing device in an AV control system that executes programmed control logic, manages device communication, and processes user input from touchpanels, sensors, and scheduling systems. The control processor communicates with AV devices via RS-232, IP, IR, relay, and other protocols, translating user interface commands	Ch. 6	control system, AV touchpanel, RS-232, IP address, relay

Term	Definition	Chapter Reference	Related Terms
	(e.g., "turn on system") into specific device commands (e.g., "power on display, switch input 2, raise volume to -10 dB, power on DSP"). Modern control processors are IP-based and managed via web interfaces and cloud portals.		
Dante (Audio Networking Standard)	A proprietary AoIP (Audio over IP) protocol and networking system developed by Audinate that enables the transmission of hundreds of channels of uncompressed, low-latency audio over standard Ethernet networks. Dante is the dominant AoIP standard in professional and educational AV, supported by hundreds of manufacturers. Dante uses standard managed network switches (without requiring AVB-capable hardware) and can share the same Ethernet infrastructure as standard IT traffic when properly configured with VLAN separation and QoS marking.	Ch. 2, Ch. 5	AoIP, AVB, AES67, network switch, VLAN, latency
dB (Decibel)	A logarithmic unit used to express the ratio of two values of power or amplitude. In audio, the decibel is used	Ch. 2	dB SPL, gain, gain structure, amplifier, frequency response

Term	Definition	Chapter Reference	Related Terms
	to express signal levels, gain, attenuation, and sound pressure levels. Because the decibel scale is logarithmic, a 3 dB increase represents approximately double the power, while a 10 dB increase is perceived by human listeners as roughly twice as loud. Decibels are always relative to a reference: dBV, dBu, and dBFS are common electrical audio references; dB SPL uses 20 μ Pa as the threshold of human hearing.		
dB SPL (Sound Pressure Level)	A measurement of the pressure of a sound wave relative to the threshold of human hearing (20 micropascals, or 0 dB SPL). Common reference levels include: whisper (~30 dB SPL), normal conversation (~65 dB SPL), classroom instruction target (70–80 dB SPL at front seats), and the threshold of pain (~120–130 dB SPL). For educational AV design, adequate speech level (minimum 65 dB SPL at the last row, with $SNR \geq 15$ dB over background noise) is essential for speech intelligibility compliance with ANSI S12.60.	Ch. 2	dB, amplifier, NC, STI, ANSI S12.60, speaker placement

Term	Definition	Chapter Reference	Related Terms
DHCP (Dynamic Host Configuration Protocol)	A network management protocol that automatically assigns IP addresses and other network configuration parameters (subnet mask, default gateway, DNS server) to devices joining a network. In educational AV, most IP-controlled devices can acquire addresses via DHCP during initial setup, but production installations should use static (manually assigned) IP addresses or DHCP reservations to ensure devices always have the same address, which is required for reliable control system communication and network monitoring.	Ch. 5	IP address, subnet mask, network switch, VLAN, control processor
Digital Signage	A network-connected display system used to present dynamic multimedia content (announcements, wayfinding, scheduling information, emergency notifications) in public and semi-public areas of educational facilities. Digital signage systems consist of display hardware, a media player, and a Content Management System (CMS) accessed via web	Ch. 9	CMS, media player, WCAG 2.1, QoS, network switch

Term	Definition	Chapter Reference	Related Terms
	browser or application. In educational AV, digital signage must comply with WCAG 2.1 AA accessibility standards including sufficient text contrast, legible font size, and alternatives for any audio or time-based content.		
DLP (Digital Light Processing)	A projection technology developed by Texas Instruments that uses a digital micromirror device (DMD) chip containing millions of microscopic mirrors to modulate light from a lamp or laser source. Single-chip DLP projectors create color through a spinning color wheel, which can cause a "rainbow effect" visible to some viewers. Three-chip DLP projectors (used in large-venue projectors) process red, green, and blue simultaneously, eliminating this artifact. DLP projectors are known for their high brightness, sharp image quality, and suitability for large-venue AV.	Ch. 3	LCD, OLED, lumen, throw ratio, projector
Document Camera	A desktop camera device designed to capture and display physical documents, textbook pages, 3D objects, and handwritten notes	Ch. 3	signal flow, HDMI, UDL, accessibility, interactive flat panel

Term	Definition	Chapter Reference	Related Terms
	on a classroom display. Document cameras output a live 1080p or higher video feed via USB or HDMI and are directly integrated into classroom AV signal chains through a switcher. In accessible classrooms, document cameras allow instructors to display materials at large scale for students with low vision and to model tactile activities for remote learners, supporting multiple means of representation consistent with UDL principles.		
DSP (Digital Signal Processor)	A programmable hardware device or software component that performs real-time digital processing of audio signals. In educational AV, the DSP is the central audio processing hub, providing functions including: equalization (EQ), acoustic echo cancellation (AEC), automatic gain control (AGC), noise suppression, signal routing, mixing, compression, limiting, and loudspeaker management. The DSP also provides dedicated outputs for assistive listening systems (FM and hearing loop), ensuring that	Ch. 2, Ch. 6	AEC, EQ, gain structure, hearing loop, ALS, Dante, amplifier

Term	Definition	Chapter Reference	Related Terms
	all users receive optimal audio regardless of their location in the room.		
Dynamic Microphone	A microphone that uses electromagnetic induction to convert sound pressure into an electrical signal. A coil of wire attached to a moving diaphragm is suspended within a magnetic field; when sound moves the diaphragm, the coil moves and generates a voltage. Dynamic microphones do not require external power, are highly durable and resistant to moisture, and handle high sound pressure levels well, making them the preferred choice for handheld wireless microphones in presentations, performances, and educational events where durability is paramount.	Ch. 2	condenser microphone, phantom power, polar pattern, wireless microphone

E - H

Term	Definition	Chapter Reference	Related Terms
EDID (Extended Display Identification Data)	A standardized data format stored in a display device's memory that communicates the display's capabilities—including	Ch. 4	HDMI, HDCP, resolution, scaler, matrix switcher

Term	Definition	Chapter Reference	Related Terms
	supported resolutions, refresh rates, color depth, and audio formats—to the connected source device. When an AV source connects via HDMI or DisplayPort, it reads the display's EDID to determine the optimal signal format to transmit. In complex AV systems with matrices, extenders, and switchers, EDID management is a critical configuration task; improper EDID handling can cause resolution mismatches, blank screens, and no-audio conditions.		
Equalization (EQ)	The process of adjusting the relative levels of specific frequency bands within an audio signal to compensate for room acoustics, microphone response characteristics, or loudspeaker frequency response anomalies. Parametric EQ allows adjustment of center frequency, bandwidth (Q), and gain for each band. Graphic EQ adjusts fixed-frequency bands. In educational AV, EQ is applied within the DSP to optimize speech intelligibility: typically boosting presence frequencies (2–5	Ch. 2	DSP, frequency response, gain structure, hearing loop, speech intelligibility

Term	Definition	Chapter Reference	Related Terms
	kHz) and attenuating problematic low-mid resonances. EQ is also used in hearing loop systems to correct for frequency response deviations per IEC 60118-4.		
Flat Panel Display	A thin, lightweight display using LCD, LED-backlit LCD, OLED, or QLED technology to produce images without the depth required by traditional CRT displays. In educational AV, flat panel displays have largely replaced projectors in small-to-medium-sized classrooms due to their day-use visibility, lower maintenance requirements (no lamp replacement), interactive touch capability, and integrated media playback. Key specifications for educational use include: size, brightness (nits), resolution (4K UHD minimum), touch capability, and input connectivity.	Ch. 3	LCD, LED display, OLED, QLED, interactive flat panel, lumen, contrast ratio
Firmware	Embedded software programmed into a hardware device (display, DSP, control processor, network switch, camera) that controls its fundamental operations and	Ch. 5, Ch. 6	control processor, network switch, DHCP, cybersecurity

Term	Definition	Chapter Reference	Related Terms
	features. Unlike user-installed applications, firmware resides in non-volatile memory and is updated by the manufacturer to fix bugs, add features, and address security vulnerabilities. In educational AV, maintaining current firmware on all networked devices is a critical cybersecurity and performance practice; outdated firmware can expose systems to known vulnerabilities and cause interoperability failures.		
FOH (Front of House)	The area of a performance venue or auditorium where the audience is seated, as distinguished from the stage or backstage areas. In audio production, "FOH" refers to the position of the main mixing console (located in the audience area) and the audio engineer operating it, whose job is to mix the sound heard by the audience. The FOH engineer controls the main PA system, monitors the overall sound quality, and coordinates with the monitor engineer (on stage) to balance the performer and audience	Ch. 2	mixing console, line array speaker, digital mixing console, gain structure

Term	Definition	Chapter Reference	Related Terms
	experiences.		
Frequency	The number of complete cycles of a sound wave or electrical signal occurring per second, measured in hertz (Hz). The human auditory range spans approximately 20 Hz to 20,000 Hz (20 kHz). In educational AV, frequency is central to understanding microphone polar patterns, loudspeaker frequency response, room acoustics, and audio equalization. Speech intelligibility depends most heavily on frequency content between approximately 500 Hz and 4,000 Hz; hearing loss frequently affects high-frequency sensitivity, making this range particularly important for accessible audio design.	Ch. 2	frequency response, equalization, dB SPL, hearing loop, STI
Frequency Response	A specification describing the range of frequencies a microphone, loudspeaker, amplifier, or other audio component can reproduce, and the relative level (in dB) at which it reproduces them across that range. Typically expressed as a range with tolerance (e.g., 80	Ch. 2	equalization, dB, DSP, amplifier, hearing loop

Term	Definition	Chapter Reference	Related Terms
	Hz–16 kHz ± 3 dB). A flat frequency response indicates the device reproduces all frequencies within the specified range at equal levels, which is generally desirable for accurate sound reproduction. Deviations in frequency response are compensated with equalization (EQ) during system commissioning.		
Gain	The amplification applied to an audio signal, expressed in decibels (dB). Gain is applied at multiple stages of an audio signal chain (preamplifier, channel strip fader, main fader, power amplifier) to raise signal levels from microphone level (~ 1 mV) to line level (~ 1 V) to loudspeaker drive level ($\sim$ tens of volts). Setting appropriate gain at each stage of the signal chain—known as gain staging or gain structure—is fundamental to achieving high signal-to-noise ratio and preventing distortion in AV systems.	Ch. 2	gain structure, dB, amplifier, DSP, signal flow
Gain Structure	The discipline of setting optimal signal levels at each stage of an audio signal chain to maximize the signal-	Ch. 2	gain, dB, DSP, amplifier, signal flow, noise suppression

Term	Definition	Chapter Reference	Related Terms
	to-noise ratio while preventing clipping (distortion). Proper gain structure begins at the microphone preamplifier (set to achieve a nominal line level output), proceeds through each processing stage (EQ, compressor, DSP) at unity gain, and ends with the power amplifier output level matched to the loudspeaker's rated sensitivity. Incorrect gain structure is the most common cause of poor audio quality, audible noise, and distortion in installed AV systems.		
Ground Loop	An electrical interference condition that occurs when two or more pieces of AV equipment are connected and have different reference ground potentials, causing unwanted alternating current (AC) to flow through the signal cable shield and introducing a low-frequency hum (typically 60 Hz in North America) into the audio signal. Ground loops are common in educational AV systems where equipment is powered from different electrical circuits. Solutions include balanced	Ch. 2, Ch. 10	balanced audio, XLR connector, troubleshooting, ground bus

Term	Definition	Chapter Reference	Related Terms
	audio connections (which reject common-mode interference), isolation transformers, and proper single-point grounding of the AV system.		
H.264 (AVC)	Advanced Video Coding, the dominant video compression standard for HD video conferencing, streaming, Blu-ray, and most online video platforms. H.264 achieves high compression efficiency (enabling 1080p video at 2–8 Mbps) while maintaining broad hardware compatibility across virtually all modern devices and platforms. In educational AV, H.264 is the de facto standard for video conferencing interoperability, lecture capture, and digital signage content delivery. Its broad support ensures compatibility between systems from different manufacturers.	Ch. 7	H.265, codec (video), bandwidth, video conferencing, streaming encoder
H.265 (HEVC)	High Efficiency Video Coding, the successor to H.264 that achieves approximately 50% better compression efficiency at equivalent visual quality. H.265 enables 4K video	Ch. 7, Ch. 12	H.264, codec (video), bandwidth, 4K, streaming encoder

Term	Definition	Chapter Reference	Related Terms
	streaming at 15–25 Mbps and 1080p at 1–4 Mbps, making it increasingly important for bandwidth-constrained educational networks and 4K AV systems. While H.265 hardware encoding and decoding support is now widespread on modern devices, it is not universally supported by all legacy conferencing platforms and endpoints, requiring careful compatibility verification during system specification.		
H.323	An ITU-T standard suite that defines protocols for audio, video, and data communications over IP networks, primarily used in traditional room-based video conferencing systems. H.323 was the dominant video conferencing protocol through the early 2000s and is still used in many legacy higher education conferencing installations. It is increasingly being superseded by SIP (Session Initiation Protocol) and cloud-based platforms (Zoom, Teams, Webex) that use proprietary or WebRTC-based protocols.	Ch. 7	SIP, video conferencing, codec (video), WebRTC, unified communications

Term	Definition	Chapter Reference	Related Terms
	Interoperability between H.323 and SIP systems requires a gateway or protocol translation device.		
HDBaseT	A connectivity standard that transmits uncompressed 4K HDMI video, audio, USB, control signals (RS-232, IR), and power (PoE) over a single CAT6A cable at distances up to 100 meters (328 feet). HDBaseT is the preferred technology for extending HDMI signals across larger educational spaces where the 5-meter practical limit of passive HDMI cables is insufficient. HDBaseT installations use a transmitter at the source and a receiver at the display, both requiring power. HDBaseT Spec 2.0 and higher support 4K/60Hz and HDR content.	Ch. 4	CAT6A, HDMI, signal extender, PoE, EDID
HDCP (High-Bandwidth Digital Content Protection)	A copy protection standard developed by Intel that encrypts digital video signals transmitted via HDMI, DisplayPort, and other digital video interfaces to prevent unauthorized copying of protected content. HDCP 2.2 is required for 4K Ultra HD content from	Ch. 4	HDMI, EDID, matrix switcher, 4K, signal integrity

Term	Definition	Chapter Reference	Related Terms
	streaming services and optical discs. In educational AV systems, HDCP compliance must be verified across the entire signal chain—source, cables, matrix switchers, extenders, and displays—as any non-compliant component in the chain will cause the video signal to be blocked entirely, resulting in a black screen.		
HDMI (High-Definition Multimedia Interface)	The dominant digital video and audio interconnection standard in consumer and prosumer AV, transmitting uncompressed digital video (up to 10K) and up to 32 channels of digital audio over a single cable. HDMI versions define supported bandwidth and capabilities: HDMI 1.4 supports 4K/30Hz; HDMI 2.0 supports 4K/60Hz at 18Gbps; HDMI 2.1 supports up to 10K and 48Gbps. In educational AV, HDMI 2.0 is the minimum recommended specification for new installations. HDMI cables have practical distance limitations of approximately 3–5 meters for passive cables at 4K/60Hz.	Ch. 4	HDCP, EDID, HDBaseT, signal extender, 4K, aspect ratio

Term	Definition	Chapter Reference	Related Terms
Hearing Loop (Induction Loop)	An assistive listening technology that transmits audio via a magnetic field created by an electrical current driven through a wire loop installed around the perimeter of a room or seating area. Hearing aid and cochlear implant users with a telecoil (T-coil) switch in their device can receive the magnetic field directly in their hearing instrument, bypassing ambient room noise entirely. Hearing loops are the only assistive listening technology that works without any additional receiver device for T-coil users. IEC 60118-4 defines the performance standard; ADA best practice strongly encourages hearing loop installation in educational assembly areas.	Ch. 2, Ch. 11	telecoil, ALS, ADA, DSP, IEC 60118-4, hearing loss
Hybrid Classroom	A learning environment designed to serve both in-room and remote participants simultaneously in real time, typically via a video conferencing platform. Hybrid classrooms require specialized AV design that addresses camera positioning, microphone coverage, speaker	Ch. 7	hybrid equity, video conferencing, PTZ camera, beamforming microphone, AEC

Term	Definition	Chapter Reference	Related Terms
	placement, and display configuration to provide equitable participation experiences for all participants regardless of their physical location. The concept of "hybrid equity" describes the goal of ensuring remote participants have equivalent access to instruction, interaction, and content as in-room students—a significantly higher design standard than simple video conferencing connectivity.		
Hybrid Equity	A design and operational principle for hybrid learning environments that holds that remote participants should have an equivalent—not merely adequate—educational experience compared to in-room students. Hybrid equity requires that remote learners can see all content (presentations, whiteboard, physical demonstrations), hear all instruction and classroom discussion clearly, participate in Q&A and discussion equally with in-room students, and access all accessibility supports	Ch. 7, Ch. 11	hybrid classroom, ISTE Standards, UDL, video conferencing, PTZ camera

Term	Definition	Chapter Reference	Related Terms
	(captioning, interpreted content) that in-room students receive. Achieving hybrid equity is a primary design driver for modern educational AV system specification.		

I - L

Term	Definition	Chapter Reference	Related Terms
Impedance	The total opposition to current flow in an AC electrical circuit, measured in ohms (Ω), encompassing both resistance and reactance. In audio systems, impedance matching between source and load devices affects signal transfer efficiency, frequency response, and noise performance. Microphones are typically 150–600 Ω (low impedance); preamplifier inputs are 1,500 Ω –10 k Ω . Loudspeakers are 4–16 Ω for low-impedance systems. In 70V constant-voltage distribution systems, transformers at each speaker tap convert impedance to allow multiple speakers to be driven from a single amplifier without loading problems.	Ch. 2, Ch. 4	balanced audio, amplifier, 70V audio, microphone, TRS connector

Term	Definition	Chapter Reference	Related Terms
Infrared (IR) Control	A control method using modulated infrared light pulses to transmit commands from a remote control to an AV device's IR receiver. IR control requires line-of-sight between the emitter and the device's receiver and does not provide confirmation feedback (one-way only). In modern installed AV systems, IR has been largely replaced by RS-232 serial and IP (Ethernet) control, which offer bidirectional communication, feedback, and status monitoring. IR emitters (wired to a control processor) are still used to control legacy or consumer-grade devices without RS-232 or IP control ports.	Ch. 6	control processor, RS-232, IP address, control system
Interactive Flat Panel (IFP)	A large-format flat panel display that combines a high-resolution 4K display with multi-touch capability (typically 20 simultaneous touch points), a built-in media player (Android OS), annotation software, wireless casting, and integrated speakers. IFPs are the dominant display technology in modern K-12 and higher education classrooms,	Ch. 3	flat panel display, touch, 4K, HDMI, ADA, EDID

Term	Definition	Chapter Reference	Related Terms
	replacing both traditional whiteboards and front-projection systems. Key specifications include: diagonal size (65"-110"), brightness (nits), touch accuracy and responsiveness, input connectivity, software ecosystem, and ADA-compliant mounting height options.		
IP Address	A numerical label assigned to each device on an IP network that identifies its location and enables communication. IPv4 addresses consist of four octets (e.g., 192.168.1.100); IPv6 uses 128-bit hexadecimal notation for a vastly larger address space. In educational AV, all networked devices (control processors, DSPs, cameras, displays, scheduling panels, signage players) require unique, stable IP addresses for reliable control system communication and remote monitoring. Static IP assignment or DHCP reservation is strongly recommended for all AV devices.	Ch. 5	DHCP, subnet mask, VLAN, network switch, control processor
ISTE Standards	The International Society for Technology in Education (ISTE)	Ch. 1, all chapters	UDL, hybrid equity, WCAG 2.1, instructional designer

Term	Definition	Chapter Reference	Related Terms
	publishes a suite of standards for learners, educators, education leaders, coaches, and computer science educators that define competencies for effective technology-integrated teaching and learning. The ISTE Standards for Educators include seven roles and associated indicators emphasizing learner-centered design, digital equity, computational thinking, collaboration, and data-driven decision-making. This textbook aligns chapter learning objectives and assessments to specific ISTE Standards indicators, providing a standards-based framework for professional development programs.		
Jitter	Variation in the arrival time of data packets in a network, measured in milliseconds (ms). In real-time audio and video communications (video conferencing, AoIP), jitter causes audio dropouts, clicks, robotic voice quality, and video artifacts when packets arrive out of sequence or too late	Ch. 5	QoS, latency, packet loss, bandwidth, video conferencing, AoIP

Term	Definition	Chapter Reference	Related Terms
	to be used by the decoder's buffer. QoS (Quality of Service) network configuration prioritizes AV traffic to minimize jitter. Acceptable jitter for video conferencing is typically less than 30 ms; AoIP (Dante) requires sub-millisecond jitter for synchronous multi-device operation.		
Latency	The delay between when a signal is captured or transmitted and when it is reproduced or received, typically measured in milliseconds (ms). In AV systems, latency affects: audio-to-video synchronization (lip sync), echo perception in reinforced audio systems, and AoIP network performance. In video conferencing, total one-way latency exceeding 150 ms becomes noticeably awkward for conversational interaction; above 300 ms causes participants to inadvertently talk over each other. In AoIP systems, Dante achieves latency as low as 0.25 ms on compatible networks.	Ch. 5, Ch. 7	jitter, QoS, AoIP, Dante, video conferencing, packet loss
Lavalier Microphone	A small, clip-on microphone designed to be worn	Ch. 2	wireless microphone, cardioid,

Term	Definition	Chapter Reference	Related Terms
	on the body (typically attached to a lapel, collar, or tie) to capture the wearer's voice at close range while remaining discreet. Lavalier (lav) microphones are typically omnidirectional or cardioid in pattern and condenser in element type, requiring phantom power or battery power. In wireless systems, the lavalier capsule connects to a bodypack transmitter worn on the body or clipped to a belt. Lavaliers provide hands-free voice capture for instructors while allowing freedom of movement throughout the classroom.		omnidirectional microphone, phantom power, bodypack
LCD (Liquid Crystal Display)	A flat panel display technology that uses a backlight (typically LED) illuminating liquid crystal cells that selectively block or pass light to create an image. LCD is the underlying technology in the majority of educational flat panel displays and most laptop displays. Key characteristics include: lower power consumption than plasma, inability to achieve true black (due to backlight bleed), good brightness in	Ch. 3	LED display, OLED, QLED, contrast ratio, flat panel display

Term	Definition	Chapter Reference	Related Terms
	ambient light conditions, and wide viewing angle (IPS panels). LCD displays using LED backlighting are typically marketed as "LED displays" in consumer and commercial markets.		
LED Display	A display technology that uses light-emitting diodes either as a backlight for an LCD panel (LED-backlit LCD, the most common consumer/commercial display type) or as the direct light-emitting pixel element (direct-view LED, used in large-format digital signage and indoor video walls). Direct-view LED displays (also called LED walls or LED video walls) offer very high brightness, wide viewing angles, modular scalability to any size, and long operational life, making them increasingly popular for auditoriums, lobbies, and large lecture halls where projection screen brightness would be insufficient.	Ch. 3	LCD, OLED, QLED, flat panel display, digital signage, DLP
Lumen (ANSI Lumen)	The standardized unit of luminous flux emitted by a projector, measured according to the ANSI IT7.228 standard. ANSI lumens are measured as the	Ch. 3	projector, throw ratio, contrast ratio, display, DLP

Term	Definition	Chapter Reference	Related Terms
	average of nine test points across the projection surface, providing a more consistent and comparable rating than manufacturer peak lumen claims. Educational classroom projectors typically range from 3,000 to 6,000 ANSI lumens for normal ambient light conditions; auditorium and large-venue projectors may require 10,000–30,000+ ANSI lumens to maintain adequate image contrast and caption legibility with room lighting on.		
LMS (Learning Management System)	A software platform that enables the delivery, management, tracking, and reporting of educational courses and learning materials. Common LMS platforms in education include Canvas, Blackboard, Moodle, D2L Brightspace, and Google Classroom. In educational AV, the LMS is the delivery platform for online resources including demo videos, assessment banks (QTI packages), downloadable templates, and interactive labs. LMS accessibility compliance (WCAG 2.1 AA minimum) is	Ch. 1, Ch. 12	QTI, WCAG 2.1, Section 508, ISTE Standards, UDL

Term	Definition	Chapter Reference	Related Terms
	required for institutions receiving federal funding under Section 508 and the Higher Education Act.		

M - O

Term	Definition	Chapter Reference	Related Terms
Matrix Switcher	An AV signal routing device that connects any input to any output simultaneously, allowing multiple sources to be displayed on multiple displays in any combination. Unlike a simple switcher (many inputs, one output) or a distribution amplifier (one input, many outputs), a matrix switcher (expressed as inputs × outputs, e.g., 4×2, 8×4) enables flexible, simultaneous routing for complex instructional environments. Modern matrix switchers are IP-controlled, HDCP 2.2 compliant, and include EDID management to ensure compatibility between all connected source and display devices.	Ch. 4, Ch. 6	HDMI, HDCP, EDID, switcher, signal flow, control processor
Microphone	An electroacoustic transducer that converts acoustic	Ch. 2	dynamic microphone, condenser

Term	Definition	Chapter Reference	Related Terms
	sound pressure into a corresponding electrical signal. Microphones differ in their operating principle (dynamic, condenser, ribbon), polar pattern (cardioid, omnidirectional, supercardioid, bidirectional), frequency response, and sensitivity. In educational AV, microphone selection is driven by the instructor's movement patterns, room acoustics, background noise levels, and integration with the AV signal chain (DSP/AEC for video conferencing). Microphones are the first and most critical link in the audio signal chain; a poor microphone choice cannot be corrected by downstream processing.		microphone, polar pattern, lavalier microphone, boundary microphone, beamforming microphone
Mixer	A device that accepts multiple audio input signals, applies gain and processing (EQ, dynamics, effects) to each channel independently, and combines them into one or more output signals. Mixers range from simple passive summing boxes to large digital mixing consoles with 96+ channels, built-in effects processors, and Dante	Ch. 2	DSP, digital mixing console, gain structure, FOH, Dante

Term	Definition	Chapter Reference	Related Terms
	networking. In educational AV, the mixing function is often integrated within the DSP for smaller installed systems. Large auditoriums and performance venues use dedicated digital mixing consoles at a front-of-house position for live event mixing.		
Multicast	A network communication method in which a single data stream is transmitted once but received by multiple destinations simultaneously, as opposed to unicast (one-to-one) or broadcast (one-to-all). Multicast is essential for AoIP (Dante, AES67) and video distribution over IP networks, where a single audio or video stream must be delivered to multiple endpoints without multiplying the network bandwidth consumed. Educational AV networks require IGMP snooping enabled on all managed switches to prevent multicast traffic from flooding all switch ports and overwhelming non-AV devices.	Ch. 5	AoIP, Dante, VLAN, managed network switch, IGMP snooping, QoS
NC (Noise Criteria)	A standardized rating system used to describe the maximum	Ch. 2	ambient noise, dB SPL, ANSI S12.60, RT60, speech intelligibility

Term	Definition	Chapter Reference	Related Terms
	acceptable background noise level in a room, based on a family of curves that account for the frequency-dependent sensitivity of human hearing. NC ratings are derived from octave-band measurements of HVAC and other mechanical noise, plotted against NC curves. NC-35 is the recommended maximum for classrooms (ASHRAE and ANSI S12.60); NC-25 to NC-30 is preferred for recording studios and critical listening environments. Spaces that exceed their NC target suffer reduced speech intelligibility and increased vocal fatigue for instructors.		
NDI (Network Device Interface)	A royalty-free video-over-IP standard developed by NewTek (now part of Vizrt) that enables live, high-quality video production workflows over standard Gigabit Ethernet networks. NDI allows cameras, computers, and production devices to share video streams across a local network without dedicated SDI cabling. In educational AV, NDI is increasingly used for PTZ camera output, streaming	Ch. 5, Ch. 7	PTZ camera, video conferencing, streaming encoder, IP address, bandwidth

Term	Definition	Chapter Reference	Related Terms
	encoder input, and production switcher integration, particularly in hybrid classrooms and campus media centers. NDI requires a Gigabit Ethernet infrastructure and can consume significant network bandwidth (100–200 Mbps per stream).		
Network Switch	A networking device that connects multiple devices within a local area network (LAN) and uses MAC address tables to intelligently forward data packets only to the appropriate destination port. Managed switches (as opposed to unmanaged switches) support configuration of VLANs, QoS, IGMP snooping, PoE, and port security—all of which are required for professional AV network integration. PoE+ (IEEE 802.3at, 30W per port) managed switches are the standard for educational AV deployments, powering IP cameras, control touchpanels, scheduling panels, and ceiling microphones from a single cable.	Ch. 5	VLAN, QoS, PoE, IGMP snooping, managed switch, AoIP, multicast
Noise Suppression	A DSP or AI-based audio processing technique that identifies and	Ch. 2	DSP, AEC, gain structure, video conferencing, NC

Term	Definition	Chapter Reference	Related Terms
	attenuates background noise (HVAC, keyboard clicks, room reverb, crowd noise) from a microphone signal while preserving speech content. Modern noise suppression algorithms use machine learning models trained on thousands of hours of speech and noise recordings to differentiate between desired and unwanted audio. In educational AV, noise suppression is particularly important in open learning environments, rooms with high HVAC noise, and video conferencing applications where background noise from participants' home environments can disrupt remote instruction.		
OLED (Organic Light-Emitting Diode)	A display technology in which each pixel generates its own light using organic compounds, eliminating the need for a separate backlight. OLED displays achieve true black (by turning individual pixels completely off), extremely high contrast ratios (often exceeding 1,000,000:1), and wide viewing angles. OLED is currently used in high-end	Ch. 3	LCD, LED display, QLED, contrast ratio, flat panel display

Term	Definition	Chapter Reference	Related Terms
	consumer televisions, professional reference monitors, and some signage applications. The primary limitations for educational AV use are higher cost, potential for image retention (burn-in) with static content, and limited peak brightness compared to high-end LCD displays used in brightly lit classrooms.		
Omnidirectional Microphone	A microphone with a spherical pickup pattern that captures sound equally from all directions (360°). Omnidirectional microphones have no proximity effect (bass boost at close range), provide consistent frequency response at all angles, and are less susceptible to handling noise than directional patterns. They are used as boundary microphones on conference tables, in measurement microphones for acoustic testing, and in some lavalier configurations. The key tradeoff is greater susceptibility to feedback in reinforced audio systems compared to cardioid or supercardioid patterns.	Ch. 2	cardioid, polar pattern, boundary microphone, lavalier microphone, gain structure

P - R

Term	Definition	Chapter Reference	Related Terms
Packet Loss	The failure of data packets to reach their destination in a network transmission, measured as a percentage of total packets sent. In real-time AV applications (video conferencing, AoIP streaming), packet loss causes audio dropouts, video freezing, pixelation, and garbled speech. Acceptable packet loss for video conferencing is typically less than 1%; AoIP networks require essentially 0% packet loss for reliable multi-channel audio. Packet loss is caused by network congestion, faulty hardware, misconfigured QoS, or inadequate bandwidth, and is diagnosed using network testing tools.	Ch. 5	QoS, jitter, latency, bandwidth, video conferencing, AoIP
Phantom Power	A method of supplying DC voltage (standardized at 48V, per IEC 61938) through microphone cables to power condenser microphone capsules and active electronics without requiring a separate power supply. Phantom power is provided by	Ch. 2	condenser microphone, XLR connector, dynamic microphone, preamplifier

Term	Definition	Chapter Reference	Related Terms
	microphone preamplifiers, mixing console channel strips, and DSP input stages. It is named "phantom" because the DC voltage is present on the microphone cable alongside the audio signal and is "invisible" to passive dynamic or ribbon microphones that do not use it. Active DI boxes and some boundary microphones also use phantom power.		
Plenum-Rated Cable	Cabling with a jacket material (typically FEP or LSZH) rated for installation in air-handling spaces (plenums)—such as the space above a dropped ceiling used as a return air pathway—without requiring conduit. Plenum-rated (CMP) cables emit significantly less toxic fumes than standard PVC-jacketed (CMR/CMX) cables in the event of a fire, reducing the risk of toxic smoke circulating through building HVAC systems. Building codes (NEC Article 800) mandate plenum-rated cables in ceiling plenum spaces; specifying the correct cable rating is a fundamental responsibility of AV system designers.	Ch. 4	CAT6A, signal extender, structured cabling, NEC, fire code

Term	Definition	Chapter Reference	Related Terms
PoE (Power over Ethernet)	A technology standard (IEEE 802.3af/at/bt) that allows network switches to deliver electrical power to connected devices over standard Ethernet cabling (CAT5e or higher), eliminating the need for a separate power supply. IEEE 802.3af (PoE) delivers 15.4W per port; IEEE 802.3at (PoE+) delivers 30W; IEEE 802.3bt (PoE++) delivers up to 90W. In educational AV, PoE+ is the standard for powering IP cameras, AV control touchpanels, room scheduling panels, ceiling microphone arrays, and other networked AV devices from the infrastructure switch.	Ch. 5	network switch, CAT6A, IP address, control processor, managed switch
Polar Pattern	A graphical representation of a microphone's directional sensitivity, showing the relative output level (in dB) at each angle of incidence for a given frequency. Common polar patterns include: omnidirectional (equal pickup in all directions), cardioid (heart-shaped, most sensitive in front), supercardioid and hypercardioid (narrower front pickup with small rear lobe), and	Ch. 2	cardioid, omnidirectional microphone, beamforming microphone, gain structure

Term	Definition	Chapter Reference	Related Terms
	bidirectional/figure-8 (equal pickup from front and rear). Polar pattern selection is a critical microphone specification decision that affects gain-before-feedback, background noise rejection, and coverage area in educational AV systems.		
PTZ Camera	A remote-controllable camera with motorized Pan (horizontal rotation), Tilt (vertical angle adjustment), and Zoom (optical magnification) capabilities. PTZ cameras are essential components of hybrid classroom and auditorium AV systems, enabling camera operators or automated AI systems to frame shots of speakers, presenters, or the room without physical camera repositioning. In educational AV, PTZ cameras support auto-framing AI (which automatically tracks active speakers), can be controlled by the AV control system via IP or RS-232, and store multiple preset positions for rapid scene switching during events.	Ch. 7	hybrid classroom, video conferencing, NDI, IP address, control processor
QoS (Quality of Service)	A set of network management	Ch. 5	VLAN, jitter, latency, packet loss,

Term	Definition	Chapter Reference	Related Terms
	techniques used to prioritize specific types of traffic (such as real-time audio and video) over less time-sensitive traffic (such as file transfers and email) to ensure consistent performance for latency-sensitive AV applications. QoS is configured on managed network switches and routers using DSCP (Differentiated Services Code Point) markings that identify packet priority levels. In educational AV networks, AoIP traffic (Dante, AES67) and video conferencing streams are marked with the highest priority (DSCP EF, value 46) to prevent packet loss and jitter under network load conditions.		managed network switch, AoIP
QLED (Quantum Light-Emitting Diode)	A display technology that uses a quantum dot filter layer to enhance the color accuracy and brightness of an LED-backlit LCD panel. Quantum dots are nanoscale semiconductor particles that emit very pure colors when illuminated by the LED backlight, resulting in a wider color gamut (wider range of reproducible colors), higher peak brightness, and better energy	Ch. 3	LCD, LED display, OLED, color temperature, flat panel display, contrast ratio

Term	Definition	Chapter Reference	Related Terms
	efficiency compared to standard LED-LCD displays. QLED displays are used in some high-end educational flat panels and commercial displays where color accuracy is important for content creation and scientific visualization applications.		
Refresh Rate	The number of times per second a display updates its image, measured in hertz (Hz). Standard educational displays operate at 60 Hz; higher refresh rates (120 Hz, 144 Hz) provide smoother motion for video content and reduce perceived flicker. Some individuals with photosensitive conditions, including certain students with autism spectrum disorder or epilepsy, may be sensitive to low refresh rates. Display specifications should note refresh rate, and educators should be aware that CRT-era projectors and some lower-cost displays may operate at 50 Hz or below, which can cause visible flicker under some lighting conditions.	Ch. 3	flat panel display, resolution, aspect ratio, LED display
Resolution	The total number of pixels displayed on a screen, expressed	Ch. 3	aspect ratio, 4K, flat panel display, HDMI, display calibration

Term	Definition	Chapter Reference	Related Terms
	as horizontal × vertical pixel counts (e.g., 1920×1080 for Full HD/1080p; 3840×2160 for 4K UHD). Higher resolution provides sharper images, finer text rendering, and greater detail visibility—all important for on-screen caption legibility, small-text documents displayed via document camera, and detailed instructional graphics. 4K UHD (3840×2160) is the current recommended minimum resolution for new interactive flat panels in educational environments, while 1080p remains acceptable for secondary confidence monitors and legacy installations.		
Reverberation	The persistence of sound in a room after the original sound source has stopped, caused by multiple reflections from room boundaries (walls, floor, ceiling). Excessive reverberation (long RT60) smears speech sounds together in time, dramatically reducing speech intelligibility and making it difficult for students—especially those with hearing	Ch. 2	RT60, NC, speech intelligibility, ANSI S12.60, acoustic conditions

Term	Definition	Chapter Reference	Related Terms
	loss or auditory processing difficulties—to understand spoken instruction. The primary treatment for excessive reverberation is the installation of sound-absorptive materials (acoustic panels, ceiling tiles, carpet) that convert acoustic energy to heat.		
RT60 (Reverberation Time)	The time (in seconds) required for a sound to decay by 60 dB after the source stops. RT60 is the primary acoustic performance metric for educational spaces; longer RT60 values indicate more reverberant rooms with reduced speech intelligibility. ANSI S12.60 sets a maximum RT60 of 0.6 seconds for classrooms with volumes up to 283 cubic meters (10,000 cubic feet), and 0.7 seconds for larger classrooms. RT60 is measured using calibrated impulse response techniques or estimated using the Sabine or Norris-Eyring equations based on room volume and surface absorption coefficients.	Ch. 2	reverberation, NC, ANSI S12.60, speech intelligibility, STI
RS-232	A serial communications standard (TIA/EIA-232) used for	Ch. 6	control processor, IP address, IR control, control system

Term	Definition	Chapter Reference	Related Terms
	bidirectional digital control communication between AV devices (displays, projectors, DSPs, matrix switchers) and control processors. RS-232 provides text-based command strings that enable control systems to power on/off, change inputs, adjust volume, and query device status. RS-232 cables are limited to approximately 15 meters (50 feet) without signal conditioning. While still widely supported in commercial AV equipment, RS-232 is increasingly supplemented or replaced by IP (Ethernet) control in modern AV systems.		

S - T

Term	Definition	Chapter Reference	Related Terms
Scaler	A hardware or software component that converts a video signal from one resolution to another (up-scaling or down-scaling) to ensure compatibility between sources and displays with different native resolutions. In educational AV	Ch. 4	resolution, aspect ratio, EDID, matrix switcher, signal extender

Term	Definition	Chapter Reference	Related Terms
	systems, scalers are often integrated into matrix switchers, signal extenders, and displays. Up-scaling converts 1080p content to 4K for display on a 4K panel; down-scaling converts 4K source output to 1080p for recording or streaming. Improper scaling can cause letterboxing, pillarboxing, or distorted aspect ratios that reduce legibility of on-screen text and captions.		
Section 504	Section 504 of the Rehabilitation Act of 1973 is a U.S. federal civil rights law that prohibits discrimination against individuals with disabilities by programs and activities receiving federal financial assistance, including public schools and most universities. Under Section 504, educational institutions are required to provide accessible AV technology and accommodations (captioning, ALS, accessible displays) and develop individualized accommodation plans for students with disabilities. Section 504 protections apply to qualified individuals with disabilities who are not covered by	Ch. 11	ADA, Section 508, assistive listening system, captioning, WCAG 2.1

Term	Definition	Chapter Reference	Related Terms
	IDEA (the Individuals with Disabilities Education Act).		
Section 508	Section 508 of the Rehabilitation Act requires federal agencies and organizations receiving federal funding to make their electronic and information technology (EIT) accessible to people with disabilities. The 2018 Section 508 Refresh incorporated the WCAG 2.0 Level AA standard as the technical baseline for web content and electronic documents. Educational institutions that receive federal funding (Title I, Title IV, research grants) are required to ensure that LMS platforms, online courseware, electronic documents, and AV technology procured with federal funds meet Section 508 standards.	Ch. 11	ADA, Section 504, WCAG 2.1, captioning, LMS, assistive listening system
Signal Extender	A device pair (transmitter and receiver) that extends AV signals—particularly HDMI—beyond the practical cable length limitations by converting the signal for transmission over CAT6A (HDBaseT), fiber optic, or coaxial cable. Signal	Ch. 4	HDBaseT, CAT6A, HDMI, fiber optic, signal flow

Term	Definition	Chapter Reference	Related Terms
	extenders are essential in educational AV installations where the display is far from the equipment rack (common in lecture halls and auditoriums) or where in-wall cabling must use the existing structured cabling infrastructure. HDBaseT-based extenders (100m over CAT6A) are the dominant technology for educational AV signal extension.		
Signal Flow	The path an AV signal takes from its point of origin (source device) through all intermediate processing, routing, and conversion devices to its final destination (display or speaker). Understanding and documenting signal flow is a fundamental AV competency required for system design, troubleshooting, and operator training. Signal flow is represented graphically in signal flow diagrams (also called block diagrams or one-line diagrams) that show devices as labeled boxes and signal paths as lines with directional arrows, annotated with connector and signal	Ch. 4	signal extender, HDMI, XLR, DSP, matrix switcher, troubleshooting

Term	Definition	Chapter Reference	Related Terms
	type. Signal flow diagrams are the primary design deliverable for educational AV system documentation.		
Signal Integrity	The ability of an AV signal to maintain its intended quality characteristics through all stages of the signal chain, from source to destination, without degradation caused by noise, attenuation, impedance mismatch, or timing errors. Signal integrity issues in digital AV systems (HDMI, HDBaseT) manifest as intermittent black screens, HDCP handshake failures, sparkle artifacts, or signal loss. In analog audio systems, signal integrity problems appear as noise, hum, signal dropout, or distortion. Proper cable selection, connector termination, and signal extender use are the primary techniques for maintaining signal integrity.	Ch. 4	HDMI, HDCP, EDID, signal extender, balanced audio, ground loop
SIP (Session Initiation Protocol)	An application-layer signaling protocol used to initiate, modify, and terminate real-time communication sessions including voice over IP (VoIP) calls and video	Ch. 7	H.323, VoIP, video conferencing, unified communications, WebRTC

Term	Definition	Chapter Reference	Related Terms
	conferencing. SIP has largely replaced H.323 as the dominant protocol for room-based video conferencing and enterprise telephony. SIP interoperability enables educational institutions to connect hardware video conferencing systems (endpoints) with cloud platforms (Zoom, Microsoft Teams, Google Meet) via SIP gateways, enabling "dial in" functionality from traditional hardware codecs to modern cloud-based collaboration platforms.		
Site Survey	A structured physical inspection and data collection process conducted before designing or proposing an AV system, used to document existing conditions including room dimensions, acoustic characteristics, lighting, existing infrastructure (power, network, cabling), existing AV equipment, and accessibility features. A thorough site survey is the foundation of accurate system design and realistic cost estimation. Conducting a site survey without commissioning a subsequent design	Ch. 8	BOM, signal flow, RT60, acoustic conditions, accessibility, structured cabling

Term	Definition	Chapter Reference	Related Terms
	eliminates surprises during installation, reduces change orders, and ensures the proposed system will perform appropriately in the actual physical environment.		
Speech Intelligibility	A measure of how well spoken words can be understood by listeners in a given acoustic environment. Speech intelligibility is affected by the signal-to-noise ratio (SNR), reverberation time (RT60), speaker-to-listener distance, microphone quality, and the listener's hearing ability. It is quantified by the Speech Transmission Index (STI) on a scale from 0 (completely unintelligible) to 1.0 (perfect intelligibility); educational AV systems should achieve $STI \geq 0.75$ ("Good" or better) at all student positions. Poor speech intelligibility disproportionately affects students with hearing loss, non-native language learners, and those with auditory processing differences.	Ch. 2	STI, RT60, NC, ANSI S12.60, dB SPL, DSP, hearing loop
STI (Speech Transmission Index)	A physically based objective measurement of speech intelligibility that quantifies how	Ch. 2	speech intelligibility, RT60, NC, ANSI S12.60, hearing loop, DSP

Term	Definition	Chapter Reference	Related Terms
	much modulation is preserved in the speech signal after transmission through a room's acoustic environment and reinforcement system. STI values range from 0 (bad) to 1.0 (excellent): 0.45–0.60 is "Fair"; 0.60–0.75 is "Good"; above 0.75 is "Excellent." STI is measured using specialized acoustic analyzers and is specified in standards including IEC 60268-16 and referenced in ANSI S12.60. In educational settings, achieving $STI \geq 0.75$ at all student positions is the recommended design target for accessible audio.		
Subnet Mask	A 32-bit number (in IPv4) that defines which portion of an IP address identifies the network and which portion identifies the host device. Subnet masks are expressed in dotted decimal notation (e.g., 255.255.255.0, or equivalently /24 in CIDR notation). In educational AV, all devices on the same AV VLAN must share the same subnet to communicate without routing. Control processors configured to send commands to AV	Ch. 5	IP address, DHCP, VLAN, network switch, control processor

Term	Definition	Chapter Reference	Related Terms
	devices must be on the same subnet or have appropriate routing configured; incorrect subnet mask assignment is a common cause of AV control system communication failures.		
Switcher	An AV device that selects one of multiple input signals and routes it to one (or more) output destinations. Unlike a matrix switcher (which routes any input to any output simultaneously), a basic switcher has one active output at a time. Switchers range from simple manual selector switches to sophisticated IP-controlled devices with built-in scalers, EDID management, and auto-switching features. In educational AV, auto-switching (where the switcher automatically selects the most recently active source) simplifies classroom operation for instructors who simply plug in their laptop and expect the room to respond automatically.	Ch. 4	matrix switcher, HDMI, EDID, signal flow, control processor
Telecoil (T-Coil)	A small electromagnetic sensor built into hearing aids and some cochlear implant processors that picks up the	Ch. 11	hearing loop, assistive listening system, ADA, IEC 60118-4, induction loop

Term	Definition	Chapter Reference	Related Terms
	magnetic field generated by a hearing loop (induction loop) system. When a user switches their hearing instrument to T-coil mode, the device receives the amplified audio signal directly from the loop as a magnetic field, bypassing the microphone entirely. This eliminates the effects of background noise, reverberation, and distance on the audio quality received by the user. The T-coil is represented by the International Symbol of Access for Hearing Loss and the letter "T."		
Throw Ratio	The ratio of the projector's throw distance (distance from lens to screen) to the screen width, used to calculate the projected image size at a given installation distance. Expressed as: $\text{Throw Ratio} = \text{Throw Distance} \div \text{Screen Width}$. A throw ratio of 1.5:1 means the projector must be 1.5 times the desired screen width away from the screen. Short-throw projectors (ratios below 1.0) and ultra-short-throw projectors (ratios below 0.4) enable closer ceiling mounting and	Ch. 3, Ch. 8	projector, lumen, DLP, display, DISCAS

Term	Definition	Chapter Reference	Related Terms
	reduce the shadow-casting effect when the instructor stands between the projector and the screen.		
TRS Connector	A cylindrical audio connector with three conductors named Tip, Ring, and Sleeve (TRS). In professional AV, 6.35mm (1/4-inch) TRS connectors carry balanced mono audio or unbalanced stereo audio. 3.5mm (1/8-inch) TRS connectors (commonly called "aux" or "headphone" connectors) carry unbalanced stereo audio and are found on consumer devices, laptops, and mobile devices. In AV installations, TRS connectors are used for balanced line-level interconnections between equipment rack components and as insert points in mixing systems. TRS connections are more susceptible to noise ingress than XLR in long cable runs.	Ch. 4	XLR connector, balanced audio, unbalanced audio, impedance, signal flow
Troubleshooting	A systematic problem-solving process used to identify, diagnose, and resolve failures or performance degradation in AV systems. Effective AV troubleshooting follows a structured	Ch. 10	signal flow, signal integrity, DSP, control system, ground loop

Term	Definition	Chapter Reference	Related Terms
	methodology: (1) gather information (what is the symptom? when did it start?); (2) establish baseline (what should the system do when working?); (3) identify most probable cause; (4) test the hypothesis; (5) implement solution; (6) verify resolution; (7) document the issue and resolution. Most AV failures fall into three categories: signal path failures (no image/audio), control failures (system won't respond), and performance issues (poor audio/video quality).		

U - Z

Term	Definition	Chapter Reference	Related Terms
UDL (Universal Design for Learning)	A research-based educational framework developed by CAST (Center for Applied Special Technology) that guides the design of flexible learning environments, materials, and assessments to meet the diverse needs of all learners from the outset. UDL is organized around three core principles: Multiple	Ch. 11, all chapters	ISTE Standards, ADA, Section 504, captioning, hearing loop, hybrid equity

Term	Definition	Chapter Reference	Related Terms
	Means of Representation (provide information in multiple formats), Multiple Means of Action and Expression (allow diverse ways to demonstrate learning), and Multiple Means of Engagement (stimulate motivation and interest). In educational AV, UDL principles inform accessibility-first design decisions including captioning, audio description, hearing loop installation, and multi-screen display configurations.		
Unbalanced Audio	An audio interconnection method using two conductors—signal and ground—that is susceptible to electromagnetic interference (EMI) and hum induction from power cables, lighting dimmers, and other noise sources, particularly over long cable runs. Common connectors for unbalanced audio include RCA (phono), TS (tip-sleeve), and 3.5mm (1/8-inch) plugs. Unbalanced connections are standard on consumer electronics but should be avoided in professional AV installations,	Ch. 2, Ch. 4	balanced audio, TRS connector, XLR connector, ground loop, impedance

Term	Definition	Chapter Reference	Related Terms
	particularly for cable runs exceeding 3–5 meters. When connecting consumer devices (laptops, tablets) to professional AV systems, a direct injection (DI) box or balanced input adapter can convert unbalanced to balanced.		
Unified Communications (UC)	An integrated platform or set of platforms that combines multiple communication and collaboration services—including video conferencing, voice calling, instant messaging, presence, file sharing, and online meetings—into a unified user experience. Major UC platforms relevant to educational AV include Microsoft Teams, Zoom, Google Meet, and Cisco Webex. Modern educational AV systems are designed to be UC-platform-agnostic (functioning with multiple UC platforms) or certified for specific platforms, ensuring compatibility with the institution's standardized collaboration environment.	Ch. 7	SIP, H.323, video conferencing, WebRTC, hybrid classroom
VLAN (Virtual Local Area Network)	A logical partitioning of a physical network into separate broadcast	Ch. 5	QoS, multicast, managed network switch, AoIP, Dante, IP address

Term	Definition	Chapter Reference	Related Terms
	domains, configured at the managed switch level, that segregates different types of network traffic without requiring separate physical cabling infrastructure. In educational AV, VLAN segmentation is used to separate AV device traffic from general campus IT traffic, preventing AV multicast streams (AoIP, IPTV) from flooding non-AV network segments and protecting AV devices from general campus network security issues. A dedicated AV VLAN with appropriate QoS markings is a standard IT/AV integration requirement for all networked AV deployments.		
VoIP (Voice over IP)	The transmission of voice communications over IP networks rather than traditional circuit-switched telephone networks, enabling voice calls via the internet or LAN infrastructure. In educational AV, VoIP capabilities integrated into DSPs and control systems enable hands-free room-to-room calling, paging to classroom speakers, and integration with campus unified	Ch. 5, Ch. 7	SIP, unified communications, video conferencing, QoS, DSP

Term	Definition	Chapter Reference	Related Terms
	communications platforms. VoIP quality is affected by the same network parameters as video conferencing: bandwidth, latency, jitter, and packet loss. SIP is the dominant signaling protocol for VoIP in educational environments.		
WCAG 2.1 (Web Content Accessibility Guidelines)	A set of internationally recognized guidelines published by the World Wide Web Consortium (W3C) Web Accessibility Initiative (WAI) that define how to make web content more accessible to people with disabilities. WCAG 2.1 is organized around four principles (Perceivable, Operable, Understandable, Robust) and three conformance levels (A, AA, AAA). Level AA is the most widely adopted target, required by Section 508 (federal technology accessibility), the EU's EN 301 549 standard, and many institutional accessibility policies. In educational AV, WCAG 2.1 AA governs the accessibility of online video content (captions, audio description), digital	Ch. 11	ADA, Section 504, Section 508, captioning, audio description, LMS

Term	Definition	Chapter Reference	Related Terms
	signage, LMS platforms, and AV control interfaces.		
WebRTC (Web Real-Time Communication)	An open-source, browser-native standard that enables real-time audio, video, and data communication directly between web browsers without requiring plugins or dedicated client software. WebRTC is the underlying technology enabling browser-based video conferencing in platforms like Google Meet, Zoom (browser version), and Microsoft Teams (web client). In educational AV, WebRTC enables any internet-connected device with a browser to join a hybrid learning session without software installation, reducing access barriers for students and guest participants who may not have institutional software licenses.	Ch. 7, Ch. 12	SIP, video conferencing, unified communications, bandwidth, latency
XLR Connector	A robust, locking, 3-pin (XLR3) or multi-pin electrical connector used universally in professional audio for microphone cables, balanced line-level interconnections, and loudspeaker connections (in some	Ch. 2, Ch. 4	balanced audio, TRS connector, phantom power, signal flow, gain structure

Term	Definition	Chapter Reference	Related Terms
	configurations). The three conductors of XLR3 carry the positive signal (pin 2), negative (inverted) signal (pin 3), and ground/shield (pin 1), enabling balanced audio transmission. XLR connectors provide a secure, locking connection that resists accidental disconnection, and the shell makes electrical contact before the signal pins during insertion, protecting sensitive equipment from voltage spikes. XLR is the de facto standard for all professional microphone and line-level audio cabling.		
Zero Latency AoIP	A design target and operational specification for AoIP (Audio over IP) networks—particularly Dante—where end-to-end audio latency is minimized to the lowest achievable value for the given network infrastructure, typically 0.25 to 1 millisecond. Zero latency (or near-zero latency) AoIP is required in applications where performers or broadcasters monitor their own audio feed in real time; latency exceeding	Ch. 2, Ch. 5	AoIP, Dante, AVB, latency, jitter, QoS, managed network switch

Term	Definition	Chapter Reference	Related Terms
	approximately 20 ms causes audible delay that disrupts natural speech and performance. Achieving minimum AoIP latency requires Gigabit Ethernet infrastructure, AVB-capable switches (for AVB), and proper QoS configuration (for Dante).		

Section E: Vendor-Neutral Procurement Checklist

This five-phase procurement checklist provides educational institutions with a comprehensive, structured process for planning, specifying, procuring, installing, and managing educational AV systems. The checklist is designed to be used by AV coordinators, technology directors, and purchasing officers. It reflects best practices from AVIXA's AV Standards and the EDUCAUSE Higher Education IT procurement guidelines. All items are vendor-neutral.

Phase 1: Needs Assessment and Planning

- Learning objectives and pedagogical needs documented for each affected space
- Stakeholder interviews completed (classroom teachers, students, IT department, administration, disability services office, facilities management)
- Existing AV equipment inventory documented for all affected spaces
- Accessibility requirements gathered from disability services office and verified against ADA, Section 504, and Section 508 obligations

- Site survey completed for all affected spaces using standardized template (see Template B-1)
- Network readiness assessment completed with IT department (bandwidth, VLAN capability, PoE capacity, managed switch availability)
- Budget range approved by administration (equipment + installation + training + contingency)
- Project timeline established with clear milestones and approval gates
- Procurement method determined (competitive RFP, formal bid, cooperative purchasing vehicle, sole source justification)
- E-Rate eligibility assessed for telecommunications and Internet access components (if applicable to institution type)
- Furniture and room preparation (construction) scope identified and coordinated with facilities team
- Faculty/instructor professional development plan initiated alongside technology planning

Phase 2: Specification Development

- Specifications are performance-based (describing required performance characteristics, not specific brand names or model numbers)
- Interoperability requirements explicitly stated (e.g., "system shall operate with UC platforms including Microsoft Teams and Zoom")
- Backward compatibility requirements documented (existing equipment to be retained and integrated)
- WCAG 2.1 Level AA accessibility compliance required in all specifications for electronic and software components
- ADA compliance required in all specifications for physical installation (control heights, accessible pathways)
- Section 508 compliance required for all technology components if project receives federal funding

- Warranty terms defined (minimum 3 years parts and labor recommended; 5 years preferred)
- Training requirements included in scope of work (minimum hours defined; user groups specified)
- Documentation requirements included in scope (as-built drawings, IP address documentation, equipment manuals, DSP program files, control system source code)
- Cybersecurity requirements documented (mandatory firmware update to current version at installation; default passwords changed; device hardening guidelines)
- Sustainability and energy efficiency requirements documented (Energy Star certification preferred; auto-standby required)
- End-of-life and e-waste disposal plan required (R2 or e-Stewards certified recycler specified)
- Preventive maintenance service agreement option included as optional bid item
- Service response time requirements defined (recommended: 4-hour response for critical failures; same-day for any assistive technology system failure)

Phase 3: Vendor and Proposal Evaluation

- Minimum three qualified vendors solicited (competitive sourcing requirement for most public institutions)
- References from at least three comparable educational institution projects verified (site visits or telephone references)
- Demonstrated experience with educational AV documented and scored in evaluation rubric
- Proof of AVIXA membership or CTS-certified technical staff requested and verified

- Live demonstration or pilot installation conducted for complex or large-scope projects
- Interoperability testing with existing campus systems completed before award
- Total Cost of Ownership (TCO) calculated for a 5-year period for each proposal (equipment + installation + maintenance + training + consumables)
- Service Level Agreement (SLA) terms reviewed and negotiated before contract execution
- Response time for critical failures defined in contract (recommended: 4-hour response; same-day for accessibility systems)
- Accessibility compliance documentation reviewed (VPAT for software products; ADA compliance statement for installed equipment)
- Vendor financial stability assessed for long-term service capability (for large-scope projects)
- Subcontractor qualifications verified if prime vendor will use specialty subcontractors

Phase 4: Installation and Commissioning

- Pre-installation meeting with integrator, IT department, and facilities conducted before any work begins
- AV VLAN configured, tested, and verified by IT team before installation begins
- Installation schedule communicated to affected faculty and scheduled outside instruction hours where possible
- Cable management plan reviewed and approved by AV coordinator before installation begins
- Power requirements verified with facilities team (dedicated circuits, UPS, load calculations)
- ADA compliance inspected at close-out (control heights, accessible pathways, signage)

- Acceptance testing criteria documented in writing before installation begins (used as basis for punch list)
- Punch list process defined (items documented, resolution timeline agreed, holdback retained until complete)
- Hearing loop field strength tested per IEC 60118-4 by qualified tester; results documented
- FM ALS system coverage verified in all areas of the room; receiver operation verified
- Final walk-through with faculty and end-users conducted before project closeout
- Training schedule confirmed and training delivered to all user groups before project closeout
- All documentation delivered (as-built drawings, IP address log, password log, DSP files, control source code, equipment manuals, warranty cards)
- Spare parts kit (recommended: 1 spare wireless microphone transmitter, spare cables, spare ALS receivers) ordered and stored on-site

Phase 5: Ongoing Management

- Preventive maintenance schedule established and documented (quarterly recommended for educational AV systems)
- Help desk support process documented and communicated to all faculty and staff users
- Firmware update policy and schedule established in coordination with IT security team
- Annual system performance review process scheduled
- Technology refresh plan established (5–7 year equipment lifecycle for major components)
- User satisfaction survey process established (annual faculty survey recommended)

- ADA/accessibility audit scheduled annually and after any system modification
- E-waste and disposal plan for replaced equipment documented (R2/e-Stewards certified recycler identified)
- Training program for new faculty and staff established (orientation-level training offered at beginning of each academic year)
- Budget planning for technology refresh cycle initiated with administration

Section F: Online Resource Portal Content Plan

F-1: 40 Demo Video Content Plan

The following 40 demonstration videos are designed to accompany the *AV Fundamentals for Educators* textbook and provide visual, hands-on reinforcement of key concepts from each of the 12 chapters. All videos must meet the accessibility requirements specified in the Accessibility Requirements column. Production should comply with WCAG 2.1 Level AA for all video content delivered via the online portal. Videos are organized by chapter alignment.

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
1	Welcome to the AV Ecosystem	Ch. 1	5 min	Identify components and stakeholders of the educational AV ecosystem	Campus walk-and-talk format; begin in lobby, move through classroom, AV closet, auditorium	Captions (98%+ accuracy) + transcript required
2	Stakeholder Roles in Educational AV	Ch. 1	4 min	Distinguish the roles of educators, AV	Interview format with 3–4 speakers representing	Captions + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
				coordinators, integrators, and IT staff	ng different stakeholder roles; brief on-screen title cards for each speaker	
3	Vendor-Neutral Evaluation Framework	Ch. 1	6 min	Apply a vendor-neutral criteria matrix to evaluate AV products and proposals	Whiteboard animation style with voiceover; show evaluation matrix being completed step by step	Captions + audio description + transcript required
4	AV Career Pathways in Education	Ch. 1	5 min	Explain the AVIXA CTS credential pathway and related career options in educational AV	Talking-head with lower-third graphics showing career ladder; include brief interview with CTS-holder	Captions + transcript required
5	Tracing an Audio Signal Chain	Ch. 2	7 min	Trace an audio signal from microphone to loudspeaker through each processing stage	Live equipment demo in installed classroom; host physically traces cable path while explaining each component; use on-screen signal flow graphic	Captions + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
					overlay	
6	Microphone Types Compared	Ch. 2	6 min	Select appropriate microphone types for educational applications based on operating principle and polar pattern	Side-by-side comparison demo using identical source audio; display audio waveforms; include SPL meter readings	Captions + transcript required
7	Measuring Sound Levels with an SPL Meter	Ch. 2	5 min	Use a sound level meter correctly and interpret SPL measurements against ANSI S12.60 standards	Hands-on measurement demo in classroom; show correct meter positioning and settings (dBA, slow response); record ambient and speech levels	Captions + transcript required
8	Setting Up an Assistive Listening System	Ch. 2	7 min	Configure and verify an FM assistive listening system and hearing loop in a classroom setting	Step-by-step hands-on setup; demonstrate connecting transmitter to DSP, distributing receivers, and testing loop field strength	Captions + audio description + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
					with a test device; include ADA signage placement	
9	Projector vs. Flat Panel: Side-by-Side	Ch. 3	6 min	Compare projection and flat panel display technologies across key performance parameters	Live side-by-side comparison in classroom with room lights on and off; use standardized test image; measure and display lux levels on screen	Captions + transcript required
10	Setting Up a Projector: Throw Ratio in Practice	Ch. 3	5 min	Calculate the required throw distance using throw ratio formula and apply it in a real installation scenario	Hands-on setup demo; show the formula on screen, measure room dimensions, calculate throw distance, position and focus projector	Captions + transcript required
11	Display Calibration Basics	Ch. 3	6 min	Calibrate display brightness, contrast, and color temperature using on-screen test patterns	Screen recording with voiceover showing calibration process on flat panel display; use PLUGE test	Captions + audio description + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
					pattern and lux meter for brightness calibration	
12	Accessibility Check: Caption Legibility on Displays	Ch. 3	4 min	Evaluate caption display quality using WCAG 2.1 contrast ratio standards and font size guidelines	WCAG contrast checker demo on screen; show compliant vs. non-compliant caption formatting ; use WebAIM contrast checker tool (free)	Captions + audio description + transcript required
13	Cable Identification Field Guide	Ch. 4	5 min	Identify common AV connectors and cables by physical appearance and describe their signal type and application	Close-up comparison video with labeled connectors laid on flat surface; hold each type up to camera and explain; use on-screen text labels	Captions + audio description + transcript required
14	Tracing Signal Flow in a Real Classroom	Ch. 4	7 min	Follow the complete signal path from source device to display and speaker in an installed classroom	Walk-through of an operational classroom; host physically traces each connection while a	Captions + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
				AV system	graphic overlay shows the matching signal flow diagram	
15	Using a Multimeter for AV Troubleshooting	Ch. 4	6 min	Use a digital multimeter safely to measure continuity, voltage, and resistance in AV cable troubleshooting	Safety-focused hands-on demo; show proper probe placement ; demonstrate testing XLR cable continuity and checking DC supply voltage; include safety warnings on screen	Captions + transcript required
16	Network Scanner Demo: Finding AV Devices	Ch. 5	6 min	Use a network scanning tool to identify and document IP-based AV devices on a local network	Screen recording with narration; use a free/open-source network scanner (e.g., Angry IP Scanner); demonstrate scanning AV VLAN and identifying device types by IP/MAC	Captions + audio description + transcript required
17	VLANs for AV: A	Ch. 5	7 min	Explain why AV	Animated diagram	Captions +

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
	Practical Introduction			VLAN segmentation is required and how it protects AV system performance	showing data flow with and without VLAN; followed by short interview with a campus IT director explaining institutional perspective	transcript required
18	Bandwidth Testing for Video Conferencing	Ch. 5	5 min	Conduct a network bandwidth and quality test and interpret results against video conferencing requirements	Live browser-based speed test demo (platform-agnostic); interpret results (download, upload, latency, jitter); compare to VC platform minimum requirements shown on screen	Captions + transcript required
19	Control System UI Tour	Ch. 6	5 min	Navigate a control system touchpanel interface and execute basic AV control functions	Screen capture of touchpanel simulator (use vendor-provided demo or open simulator); walk through powering system on/off,	Captions + audio description + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
					source selection, volume control, and help function	
20	Control System Programming Concepts	Ch. 6	7 min	Understand event-driven control logic and the relationship between triggers, conditions, and actions	Whiteboard animation; build a simple event-driven logic flowchart step by step; use generic pseudocode, not a specific vendor language	Captions + transcript required
21	Accessibility Evaluation of a Control Interface	Ch. 6	6 min	Apply WCAG 2.1 AA criteria to evaluate the accessibility of an AV touchpanel user interface	Side-by-side comparison of accessible (large text, high contrast, clear labels) vs. inaccessible (small text, low contrast, icon-only labels) touchpanel UI designs; use WCAG contrast checker	Captions + audio description + transcript required
22	Video Conferencing Room Setup: Step by	Ch. 7	7 min	Configure a basic room-based video	Live setup demo in hybrid classroom; connect	Captions + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
	Step			conferencing system from hardware connection to test call	camera, audio bar, and display; configure platform settings; place and evaluate test call with remote participant	
23	Hybrid Equity Audit in Action	Ch. 7	6 min	Conduct a hybrid equity assessment of a classroom AV system from the perspective of both in-room and remote participants	Walkthrough of hybrid classroom; simultaneously show in-room view and split-screen remote participant view; use checklist on-screen; demonstrate both compliant and non-compliant configurations	Captions + audio description + transcript required
24	Content Sharing Best Practices	Ch. 7	5 min	Share content effectively in a hybrid meeting context using multiple methods (screen share, wireless, physical connection)	Screen recording demo; show three content sharing methods; demonstrate accessibility considerations (resolution, caption	Captions + audio description + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
					passthrough, system audio sharing)	
25	Enabling and Customizing Captioning	Ch. 7	5 min	Configure automatic speech recognition (ASR) captions and evaluate when CART escalation is required	Platform demo (platform-agnostic, show steps applicable to multiple platforms) ; demonstrate caption on/off, font size customization, speaker identification; evaluate ASR accuracy with and without proper microphone use	Captions + audio description + transcript required
26	Site Survey Walkthrough	Ch. 8	7 min	Conduct a complete AV site survey using the standardized survey template	On-location walkthrough of a classroom; host completes each section of the survey form (B-1) in real time; use on-screen graphic to highlight current survey section	Captions + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
					being completed	
27	Calculating Throw Ratio and Display Size	Ch. 8	5 min	Apply the DISCAS formula and throw ratio calculation to determine optimal display size and projector placement	On-screen calculation demo using annotation ; work through DISCAS (screen height = farthest seat ÷ 4) and throw ratio calculations step by step with classroom example measurements	Captions + audio description + transcript required
28	AV Design in draw.io: A Starter Tutorial	Ch. 8	7 min	Create a basic AV signal flow diagram using free draw.io diagramming software	Screen recording tutorial; use draw.io (free, browser-based); demonstrate creating a basic classroom signal flow diagram from scratch using the C-1 example from this appendix	Captions + audio description + transcript required
29	Digital Signage Setup: Hardware to CMS	Ch. 9	7 min	Configure a digital signage media player and connect it	Live hardware demo; show media player	Captions + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
				to a content management system for content scheduling	hardware, network configuration, and CMS web interface; schedule and deploy test content; verify display output	
30	Content Management System Walkthrough	Ch. 9	6 min	Navigate a digital signage CMS interface to upload content, create a schedule, and publish to displays	Screen recording of CMS interface; use a free-tier or demo CMS account; show content upload, template selection, scheduling, and zone configuration	Captions + audio description + transcript required
31	Accessibility Audit: Digital Signage	Ch. 9	5 min	Apply WCAG 2.1 contrast and text size criteria to evaluate digital signage content for accessibility compliance	Contrast checker demo on signage template examples; show compliant vs. non-compliant designs; demonstrate font size measurement and contrast ratio check	Captions + audio description + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
32	AV Troubleshooting Methodology	Ch. 10	6 min	Apply a systematic, seven-step troubleshooting methodology to diagnose an AV system failure	Animated troubleshooting flowchart followed by live demo applying the methodology to a simulated "no signal" failure; narrate each step of the process	Captions + transcript required
33	Diagnosing Common Audio Faults	Ch. 10	7 min	Identify and resolve common audio system problems including hum, noise, dropout, and no-audio conditions	Live fault simulation and repair; intentionally create each fault (ground loop hum, loose connector dropout, gain structure clipping); diagnose and repair each one on camera	Captions + transcript required
34	Diagnosing Common Video Faults	Ch. 10	7 min	Identify and resolve common video system problems including no signal, wrong resolution, HDCP errors, and color	Live fault simulation and repair; simulate HDCP handshake failure, EDID mismatch, and resolution scaling error;	Captions + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
				issues	diagnose each using systematic approach	
35	AV Accessibility Audit Walkthrough	Ch. 11	7 min	Conduct a complete AV accessibility audit of a classroom using the AV Accessibility Audit Tracker template	On-location audit using the audit tracker spreadsheet (Template F-2.7); check all ADA-required elements; narrate findings; demonstrate both compliant and non-compliant examples	Captions + audio description + transcript required
36	Testing a Hearing Loop System	Ch. 11	6 min	Verify hearing loop field strength across a room using a loop tester and interpret results against IEC 60118-4	Hands-on measurement demo in looped room; use a loop field strength meter; measure at multiple points; record results on measurement log (Template F-2.1); compare to IEC 60118-4 target (−3 to +3 dB re 100 mA/m)	Captions + audio description + transcript required

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
37	Captioning Setup: From ASR to CART	Ch. 11	5 min	Configure ASR-based captions, evaluate their quality, and implement CART as an escalation option	Platform comparison demo (platform-agnostic); demonstrate ASR accuracy with clear vs. accented vs. multi-speaker audio; contrast with CART output quality and setup workflow	Captions + audio description + transcript required
38	XR in the Classroom : A Demo	Ch. 12	6 min	Explore educational applications of extended reality (XR) including VR, AR, and MR in classroom settings	Campus XR lab demo; show hardware setup, student interaction , and content examples; interview instructor using XR in curriculum ; address accessibility considerations (headset adjustments for glasses users, motion sickness)	Captions + audio description + transcript required
39	AI-Powered AV Rooms	Ch. 12	5 min	Explore AI-driven features in	Live AI room demo;	Captions + transcript

Video #	Title	Chapter	Est. Duration	Key Learning Objective	Production Notes	Accessibility Requirements
				modern AV rooms including auto-framing cameras, noise suppression, and smart scheduling	show camera auto-framing tracking a moving presenter; demonstrate AI noise suppression before/after comparison; show occupancy-based auto-scheduling	required
40	AV Career Pathways: Planning Your Next Step	Ch. 12	5 min	Map a personal AV career development pathway using AVIXA credentials, ISTE certifications, and professional development resources	Talking-head with on-screen career map graphics; include brief interviews with practitioners at different career stages; show AVIXA CTS credential pathway clearly	Captions + transcript required

F-2: Downloadable Measurement Spreadsheets (8 Templates)

The following eight measurement and tracking spreadsheet templates are available for download from the online resource portal. Each template is provided in .xlsx format with unlocked cells for data entry and locked header rows, formulas, and

formatting. All templates include an instruction tab and a sample completed row for reference.

#	Template Name	Purpose	Key Columns / Fields
1	SPL Measurement Log	Documents sound pressure level measurements in classroom and auditorium environments for compliance with ANSI S12.60	Date Room ID Microphone Type Measurement Position Background Noise Level (dBA) Speech/Program Level (dBA) SNR Calculation (dB) Standard Reference Pass/Fail (auto-calculated) Notes Surveyor Name
2	Lux / Footcandle Measurement Log	Documents ambient light measurements to verify display brightness adequacy in educational spaces	Date Room ID Lighting Condition (all on/partial/off) Measurement Position Measured Lux Converted Footcandles (auto-calc) Standard Reference Display Brightness Recommendation Pass/Fail Notes
3	Network Bandwidth and Quality Test Log	Documents network performance data for AV and video conferencing applications at multiple test points across campus	Date Building/Room Device Tested IP Address VLAN Download Speed (Mbps) Upload Speed (Mbps) Latency (ms) Jitter (ms) Packet Loss (%) VC Minimum Requirement Pass/Fail (auto-calc) Notes
4	Cable Continuity Test Log	Documents structured cabling and AV cable test results for project acceptance and troubleshooting documentation	Cable ID Cable Type (CAT6A/HDMI/XLR/TRS) Length (ft) Pin 1 Continuity Pin 2 Pin 3 Pin 4 Pin 5 Pin 6 Pin 7 Pin 8 Shield Continuity Short/Cross Test

#	Template Name	Purpose	Key Columns / Fields
			Pass/Fail (auto-calc) Notes Test Date Technician
5	AV Equipment Inventory Tracker	Maintains a comprehensive asset register of all AV equipment across campus for maintenance, budgeting, and lifecycle planning	Asset ID Building Room Device Type Manufacturer Model Serial Number Purchase Date Purchase Cost Warranty Expiration Current Firmware Last PM Date Condition (1-5) Replacement Year (calc) Notes
6	Preventive Maintenance Schedule	Tracks scheduled and completed preventive maintenance tasks for all AV equipment across campus on a monthly basis	Device ID Device Type Location PM Task Description Frequency (Monthly/Quarterly/Annual) Assigned Technician Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Annual Completion % (auto-calc) Notes
7	AV Accessibility Audit Tracker	Documents ADA and WCAG 2.1 compliance status for each AV-equipped room and tracks remediation progress	Building Room ID ADA/WCAG Requirement Regulation Reference Current Status (Compliant/Non-Compliant/N-A) Priority (High/Med/Low) Estimated Remediation Cost Responsible Party Target Completion Date Date Resolved Resolution Notes
8	Project Budget Tracker	Tracks budget, vendor quotes, purchase orders, and actual expenditures for AV installation projects	Category Item Description Quantity Unit Vendor Quote 1 Vendor Quote 2 Vendor Quote 3

#	Template Name	Purpose	Key Columns / Fields
			Lowest Quote (auto) Selected Vendor PO Number PO Date Actual Cost Variance (auto-calc) Notes

F-3: LMS Assessment Package (QTI 2.1)

The LMS Assessment Package provides a complete, standards-aligned question bank of 120 items in QTI 2.1 format, suitable for import into all major learning management systems used in education. The package supports both formative (quick check) and summative (practice exam) assessment delivery.

Package Contents

Component	Description	Item Count
Full Practice Exam Bank	100 multiple-choice and multiple-select questions covering all 12 chapters in proportional distribution; designed to simulate the AVIXA AV Technologist written assessment format	100 items
Quick-Check Question Bank	20 short, chapter-level formative assessment questions (1-2 per chapter) for low-stakes comprehension checking at the end of each lesson	20 items
Total Question Bank	All 120 items with full metadata tags (see below)	120 items

Question Metadata Schema

Every question in the QTI package is tagged with the following metadata fields, enabling LMS instructors to filter, sort, and assign questions by any combination of criteria:

Metadata Field	Values / Description
Question ID	Unique identifier in format: AVFE_CH##_Q### (e.g., AVFE_CH02_Q007)
Chapter	Ch. 1 through Ch. 12
AVIXA CTS Domain	Mapped to the AVIXA AV Technologist knowledge domain(s) as published in the current AV Technologist Handbook
ISTE Standard	Aligned to specific ISTE Standards indicator (e.g., ISTE Educator 5.b)
Bloom's Taxonomy Level	Remember / Understand / Apply / Analyze / Evaluate / Create
Difficulty Level	Easy / Medium / Hard (based on cognitive demand and prerequisite knowledge required)
Question Type	Multiple Choice (single correct) / Multiple Select (multiple correct) / True-False
Accessibility Notes	Alt text for any image elements; reading level estimate (FK Grade 10 target)

QTI Package Folder Structure

/QTI_Package/	imsmanifest.xml	(Package manifest; required for LMS import)
/questions/	ch01/	(Chapters 1–12, 8–12 items each)
ch02/	ch03/	ch04/
ch05/	ch06/	ch07/
ch08/	ch09/	ch10/
ch11/	ch12/	/assessments/
full_practice_exam.xml	(100-item randomized exam configuration)	quick_check_bank.xml
(20-item chapter-level quick check pool)	/metadata/	standards_alignment.csv
(Full mapping of all 120 items to ISTE + AVIXA domains)	question_index.csv	
(Master list of all items with metadata fields)	/media/	/images/
(Alt-text-tagged images referenced in questions)		

LMS Import Instructions

LMS Platform	Import Path	Notes
Canvas (Instructure)	Settings → Import Course Content → QTI .zip file → Select file → Import	Canvas supports QTI 2.1; import as "QTI Quizzes" type; questions appear in Question Banks
Blackboard	Course Tools → Tests, Pools and Surveys → Question Pools → Import → Select QTI 2.1 package	Blackboard Ultra may have slightly different path; check platform version; some formatting may require review after import
Moodle	Question bank → Import → File format: IMS QTI 2.1 → Upload file → Import	Moodle 3.x and above support QTI 2.1; verify version before import; use "Question bank" import, not "Course restore"
Google Classroom	Native QTI import not supported. Workaround: Use Google Forms (manual re-entry) or third-party integration tool such as Formative or Edulastic, which support QTI import and Google Classroom assignment integration.	Recommend using a QTI-compatible third-party quiz tool that integrates with Google Classroom via LTI
D2L Brightspace	Edit Course → Import/Export/Copy Components → Import Components → Select "Question Library" → Upload QTI zip file	D2L supports QTI 2.1; imported questions appear in Question Library; create quizzes from library

Accessibility Note - QTI Package

All 120 questions in the QTI package comply with the following accessibility standards: (1) All image-based question elements include descriptive alt text; (2) Question stems use plain language targeting a Flesch-Kincaid Grade 10 reading level; (3) The QTI package has been tested with JAWS screen reader (Windows) and VoiceOver (macOS) in Canvas and Moodle environments; (4) No time limits are set as defaults—instructors should consult with their institution's disability services office before enabling time limits for any assessment delivery. Contact the editorial team at [accessibility@placeholder.org] for accommodation format

requests.

PART TWO: PRODUCTION TIMELINE AND PROJECT GUIDE

Part Two is directed primarily at the editorial production team at ISTE and AVIXA, contributing authors, subject matter expert reviewers, and the project manager responsible for coordinating development of the *AV Fundamentals for Educators* textbook package. It provides a phased production timeline, reviewer guidance, budget framework, and legal/licensing framework for the complete publication project.

Section G: 8–12 Week Production Timeline

The following Gantt-style timeline organizes the production of the complete *AV Fundamentals for Educators* textbook package into eight sequential phases spanning 12 weeks of active development, plus a post-delivery pilot phase. The timeline assumes all stakeholders are available as planned and that first-draft writing begins no later than Week 2. Phase overlaps are shown where parallel workstreams are possible.

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
Phase 1 - Project Launch	Weeks 1-2	Kickoff meeting with ISTE and AVIXA	Signed project charter; approved production	Project Manager (ISTE/AVIXA joint)	Stakeholder availability ; budget approval	All parties have signed the project charter;

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		editorial directors Confirm and lock chapter list and full scope Finalize SME review panel and secure written commitments Set up shared cloud workspace with folder structure and access permissions Establish and communicate file naming conventions and version control process Review and approve this production timeline Assign writing responsibilities per chapter with	timeline; confirmed SME roster		from both organizations	shared workspace is live with all contributors having verified access; SME commitments received in writing

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		deadlines Establish communication cadence (weekly check-in meeting schedule)				
Phase 2 – Content Development	Weeks 2-5	Textbook manuscript — all 12 chapters, first draft Instructor Guide first draft (all 12 chapters) Lab Workbook first draft (all 12 lab exercises)) Practice Exam first draft (100 questions with answer explanations) Resource Appendix (this document) first draft Slide Deck first draft (all 12 chapter decks, consistent	All six first-draft documents uploaded to shared workspace ; milestone progress confirmed weekly	Lead Author(s) + Contributing Authors (per chapter assignments)	Project charter approved (Phase 1); SME panel confirmed; writing assignments distributed with deadlines	All six documents present in shared workspace as first-draft files (complete, not placeholder); PM confirms receipt and readiness for Phase 3

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		t template) Weekly word-count and milestone check-in with PM				
Phase 3 – Internal Review	Weeks 5-6	Lead author self-review all six documents for internal consistency Cross-reference all ISTE and AVIXA standards alignments against chapter content Vendor-neutrality scan: verify no brand names in body text of any instructional document Accessibility and equity language review (Equity Spotlight callouts,	Internally reviewed drafts (Version 1 – Internal) uploaded; SME review package assembled and ready for distribution	Lead Author + Copyeditor	All first drafts complete (Phase 2)	Lead author signs off on internal review completion; copyeditor confirms initial pass complete; PM approves distribution to SME panel

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		UDL references, inclusive examples) Copyedit for tone, reading level, and consistency (Flesch-Kincaid Grade 10-12 target for textbook) Resolve content gaps or overlaps between documents identified during review Prepare SME review instructions and review matrix template				
Phase 4 - SME Review	Weeks 7-8	AVIXA CTS-certified technical reviewer reviews all 12 chapters for AV accuracy, BOM template	Consolidated SME Feedback Matrix (all reviewer comments merged and prioritized)	Project Manager (collection and consolidation); SME Panel (review)	Internally reviewed drafts distributed to SMEs with clear review instructions and template; all SME agreements	All six SME reviewers have submitted completed feedback; PM has consolidated into revision matrix; conflicting feedback

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		s, signal flow diagrams, and glossary definitions ISTE Standards Specialist verifies all ISTE indicator alignments in every chapter K-12 classroom teacher pilot (2-3 teachers) reviews Chapters 1, 7, and 11 for practical relevance UDL/Accessibility Specialist reviews Chapter 11, all Equity Spotlights, and Lab 11 Instructional Designer reviews Instructor Guide structure and Lab Workbook			executed	items flagged for author decision

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		pedagogy AVIXA Credentia ling SME reviews Practice Exam for domain alignmen t and item quality PM collects all feedback into unified revision matrix within 3 business days of SME submissio n deadline				
Phase 5 - Revision Round 1	Week 9	Author incorpora tes all non- conflictin g SME feedback into all six documen ts PM facilitates resolution of conflictin g feedback (author makes final call with PM	Revised document s (Version 2) ready for accessibili ty review; completed revision matrix showing all SME items resolved or deferred with rationale	Lead Author (primary); PM (conflict resolution facilitation)	Consolidat ed SME Feedback Matrix complete (Phase 4)	Lead author confirms all items in revision matrix addressed ; PM confirms Version 2 files uploaded to shared workspace ; revision matrix fully resolved

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		sign-off) Standard s alignmen t tables updated if SME identifies uncovere d indicators Slide decks and Lab Workboo k revised to match any content changes in textbook manuscri pt Practice Exam revised based on credentia ling SME feedback; answer explanati ons updated as needed Revision matrix updated to "resolved " status for each item				
Phase 6 - Accessib	Week 10	WCAG 2.1 AA audit of	Accessibili ty Audit Report	Accessibili ty Specialist	Version 2 document s	Accessibili ty specialist

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
Accessibility and Equity Review		all electronic documents (PDF, DOCX, PPTX) by accessibility specialist UDL alignment check across all six documents Equity language and representation review: inclusive example names, diverse scenarios , gender-neutral language Hearing loop and captioning content accuracy verified by assistive technology specialist Document structure review: heading hierarchy , alt text guidance for	with itemized required revisions; sign-off checklist from accessibility specialist	(primary); Lead Author (response to findings)	complete (Phase 5)	submits signed audit report; all critical and high-priority findings documented; author confirms awareness of required revisions

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		images, color contrast in slide decks (minimum 4.5:1 ratio for text) Deaf/Hard of Hearing community reviewer verifies lived-experience accuracy of accessibility content				
Phase 7 - Final Production	Weeks 11-12	Incorporate all accessibility revision feedback into all six documents Professional final copyedit (full pass across all six documents) Document design and layout: professional	Complete production-ready file package: all six documents in all required formats; QTI package tested; spreadsheet templates validated	Designer / Production Coordinator (layout and format); Lead Author (final content sign-off); PM (checklist and delivery coordination)	All revisions from accessibility review incorporated and confirmed; final copyedit manuscript provided to designer	PM conducts final deliverables checklist against project charter scope; all files present in final naming convention; accessibility specialist confirms PDF/UA compliance; QTI import tested successfully

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		nal typesetting for textbook (InDesign or equivalent) Slide deck final design: consistent branded template, ISTE/AVIX A co-branding applied per brand guidelines File format conversion: DOCX (editable instructor version), PDF/UA (accessible PDF, tagged), HTML5 (online portal version) QTI package final build: all 120 questions coded, metadata tagged, and packaged ; tested in Canvas, Blackboard				y in at least two LMS platforms

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		d, and Moodle Measurement spreadsheet templates finalized and tested in Microsoft Excel and Google Sheets Final file naming, versioning, and archiving per agreed conventions All deliverables cross-checked against project charter scope list				
Phase 8 – Delivery and Pilot	Weeks 12+ (post-delivery)	Deliver complete file package to ISTE and AVIXA editorial directors for joint review and formal acceptance Pilot course launch at	Formal delivery acceptance from ISTE and AVIXA; pilot feedback report; errata process live	Project Manager (delivery); Pilot Institution Faculty (pilot delivery); ISTE/AVIXA Editorial Directors (acceptance)	Final production file package complete and delivered (Phase 7)	ISTE and AVIXA editorial directors provide written acceptance confirmation; pilot institution confirms receipt and course setup; errata submission email

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		one or two volunteer institutions (optional, strongly recommended) Collect pilot feedback via faculty survey (end of pilot week 4) and student survey (end of pilot) Compile pilot feedback report with prioritized revision recommendations Plan Version 1.1 revision cycle (target: 6 months post-launch) based on pilot data Establish public-facing errata process and log				address active and monitored

Phase	Week(s)	Key Tasks	Primary Deliverable	Owner Role	Dependencies	Completion Criteria
		(hosted on ISTE/AVIX A joint webpage) Archive all working files, source documents, and version history in institutional repository				

Timeline Risk Factors

The following risks have the highest probability of causing schedule delays and should be actively monitored by the Project Manager: (1) **SME availability:** SME reviewers are typically working professionals with competing commitments; a 2-week reminder schedule and a firm submission deadline with a defined consequence (PM assigns a replacement reviewer) are recommended. (2) **First-draft completeness:** Incomplete or placeholder first drafts submitted at the Phase 2 deadline will compress internal review time; weekly milestone check-ins during Weeks 2-5 are essential. (3) **Accessibility remediation scope:** The accessibility audit (Phase 6) frequently surfaces more revisions than anticipated; allocate a 3-business-day buffer within Week 10 for author response.

Section H: Recommended SME Reviewers

The following table describes the nine recommended Subject Matter Expert (SME) reviewer roles for the *AV Fundamentals for Educators* textbook package. These roles are defined by expertise area rather than specific individuals; the Project Manager and ISTE/AVIXA editorial teams are responsible for identifying and engaging specific candidates. Where possible, reviewers should represent diverse institutional contexts (urban/suburban/rural, K-12/higher education, public/private) and diverse personal backgrounds to support the textbook's equity and inclusion goals.

Role	Expertise Area	Organization Type	Review Scope	Engagement Type	Est. Hours	Notes
CTS-Certified AV Design Engineer (K-12 / Higher Ed experience)	AV system design, signal flow, networking, technical specifications, BOM development, site survey methodology	AVIXA corporate member firm (integrator) or independent consultant with verifiable educational AV portfolio	Full manuscript technical accuracy review; all BOM templates (Section A); signal flow diagrams (Section C); glossary definitions (Section D); procurement checklist (Section E)	Contracted paid review; independent contractor agreement required	20-30 hours	Prefer CTS-D (design) credential; require references from at least 2 completed educational AV projects; conflict of interest disclosure required
AVIXA Credentialing Subject Matter Expert	AVIXA AV Technology test domains, item writing standards, knowledge taxonomy, and credentialing exam development methodology	AVIXA staff member or AVIXA-approved item writing pool participant	Practice exam (100 questions): content accuracy, domain alignment, item quality, distractor validity, and bias review; standards alignment CSV in QTI package	AVIXA-facilitated engagement; AVIXA to coordinate internally and provide review feedback through established process	10-15 hours	AVIXA editorial director to initiate and manage this engagement; exam content subject to AVIXA item security protocols

Role	Expertise Area	Organization Type	Review Scope	Engagement Type	Est. Hours	Notes
ISTE Standards Specialist	ISTE Standards for Educators indicators, applications in professional development, standards-based curriculum design, and digital equity frameworks	ISTE staff member, ISTE Certified Educator, or university faculty with published work in ISTE Standards application	Standards alignment verification across all 12 chapters; review of all ISTE Standards call-out boxes and standards alignment tables; Instructor Guide standards mapping; QTI metadata ISTE fields	ISTE-facilitated engagement; ISTE editorial director to coordinate	15–20 hours	ISTE editorial director to initiate; reviewer should have direct familiarity with 2023 ISTE Standards revision process
Universal Design for Learning (UDL) Specialist	UDL framework (CAST), inclusive design principles, accessibility-first content development, disability equity, and neurodiversity in educational design	CAST (Center for Applied Special Technology), university disability center, educational nonprofit with UDL focus, or CAST-trained UDL implementation specialist	Chapter 11 (comprehensive); all Equity Spotlight callout boxes across all 12 chapters; Lab 11 (accessibility lab exercise); Instructor Guide differentiation strategies; Section D glossary entries for ADA, UDL, Section 504, Section 508, WCAG 2.1	Contracted paid review; independent contractor agreement required	10–15 hours	Prioritize reviewers with both UDL expertise and lived disability experience; provide CAST's UDL Guidelines 3.0 as reference framework for review
K-12 Classroom Teacher	Teacher perspective,	Active classroom teacher in	Full manuscript	Volunteer with profession	8–12 hours per teacher;	Recruit through ISTE

Role	Expertise Area	Organization Type	Review Scope	Engagement Type	Est. Hours	Notes
with AV/Technology Experience	practical classroom technology application, instructional workflow, student behavior patterns, and realistic institutional constraint awareness	a K-12 school district; recruit to represent urban, suburban, and rural contexts (one teacher each)	readability review from teacher perspective; Lab Workbook practical feasibility (Labs 1–3); Chapter 7 (hybrid classroom applicability); Instructor Guide usability; all "Educator Spotlight" features	al development credit and/or stipend (\$200–\$500 per teacher recommended)	recruit 3 teachers	educator network and/or state technology directors associations; provide simple one-page review guide; avoid overloading with technical review tasks outside pedagogical focus
Higher Education Instructional Designer	Curriculum mapping, backward design, LMS integration, assessment development (Bloom's taxonomy alignment), and online/hybrid course design	University center for teaching and learning, instructional design department, or community college instruction office	Instructor Guide structural review; Assessment Bank (QTI) item quality and Bloom's taxonomy alignment; Lab Workbook learning objective alignment; Section F-3 (QTI package) LMS import verification	Contracted paid review; may be facilitated through university partnership agreement	10–15 hours	Request candidate with experience in at least two of the five LMS platforms in the QTI import instructions; prior course design for AV or technology topics preferred but not required
ADA / Section 508 Compliance Specialist	Legal accessibility requirements, ADA Title II and	Federal or state government agency (DOE, DOJ),	Chapter 11 (legal accuracy of all ADA, Section 504, and	Contracted paid review; may require attorney	8–12 hours	This reviewer focuses specifically on legal accuracy

Role	Expertise Area	Organization Type	Review Scope	Engagement Type	Est. Hours	Notes
	III compliance, OCR complaint resolution, Section 508 technical standards, WCAG 2.1 legal application, and institutional accessibility policy	disability rights law firm, or educational nonprofit specializing in ADA compliance and technical assistance	Section 508 references); procurement checklist accessibility phases (Section E); site survey accessibility checklist (Template B-1); all BOM templates (ADA notes column)	engagement for legal accuracy review; independent contractor agreement required		of regulatory language; not responsible for technical AV content. Coordinate with ISTE and AVIXA legal counsel regarding any specific compliance claims before publication.
Campus IT Director with AV / Network Experience	AV networking, IT/AV collaboration and integration, campus network architecture, cybersecurity policy, and institutional IT procurement perspective	K-12 technology director or higher education campus IT director with verifiable AV network integration experience; EDUCAUSE member preferred	Chapter 5 (networking) technical accuracy; all network-related content in signal flow diagrams (Section C); VLAN configuration content; procurement checklist Phase 1 network readiness and Phase 5 cybersecurity items; QTI metadata network domain questions	Volunteer engagement with acknowledgment in published textbook; no fee required but budget for one courtesy institutional copy of published textbook	5–8 hours	Conduct initial review as structured interview (60–90 min) followed by written review of targeted sections; EDUCAUSE and CoSN networks are good sources for candidate identification

Role	Expertise Area	Organization Type	Review Scope	Engagement Type	Est. Hours	Notes
Deaf / Hard of Hearing Educator or Disability Advocate	Lived experience with hearing-related disability; personal use of ALS systems, hearing loops, captioning, and other AV accessibility accommodations in educational settings; advocacy perspective	Disability advocacy organization, school for the Deaf, state commission for the Deaf and Hard of Hearing, or post-secondary disability services office	Chapter 11 (accuracy and tone of all hearing accessibility content); all assistive listening system and hearing loop content across all chapters; captioning content (Chapters 7 and 11); the F-2.7 AV Accessibility Audit Tracker template; Video #8, #36, and #37 production guidance	Compensated community review; rate should reflect expertise and lived experience (\$75-\$125/hour recommended); community partnership agreement rather than standard contractor agreement	4-6 hours	This reviewer's role is to verify that the textbook's treatment of Deaf/HoH access is accurate, respectful, and reflects community-centered language norms (e.g., Deaf with capital D when referring to Deaf culture/community); provide reviewer with the NAD style guide as a reference resource

Conflict of Interest Policy

All SME reviewers must disclose any financial relationship with AV manufacturers, distributors, or vendors before beginning their review. Reviewers with undisclosed commercial relationships may be disqualified. The vendor-neutral commitment of this textbook requires that no reviewer's feedback be used to promote, favor, or disparage any specific brand or product. All SME agreements should include a standard conflict of interest disclosure and attestation clause. Contact the ISTE or AVIXA legal team for the current standard reviewer agreement template.

Section I: Budget Estimate (Illustrative Ranges)

The following budget table provides illustrative cost ranges for the full production of the *AV Fundamentals for Educators* textbook package. These figures are intended as planning estimates only and should be refined with actual vendor quotes, contract negotiations, and institutional resource availability before being used for formal budget approval. Ranges reflect the difference between leveraging institutional resources and in-house talent (low end) versus full external professional engagement (high end). The most significant variable is video production model, which alone accounts for a \$80,000 range.

Budget Category	Low Estimate	High Estimate	Notes and Assumptions
SME Reviews (9 reviewer roles, variable hours per role)	\$8,000	\$18,000	Low: some reviewers volunteer (K-12 teachers, IT director, ISTE/AVIXA staff); High: all reviewers contracted at professional rates. Includes stipends, honoraria, and contracted review fees. Does not include ISTE or AVIXA staff time charged internally.
Lead Author(s) Fees	\$25,000	\$50,000	Low: single lead author with institutional affiliation support; High: shared lead authorship with two co-authors splitting responsibility. Includes textbook manuscript, Instructor Guide, and Lab Workbook authorship. Does not include contributing

Budget Category	Low Estimate	High Estimate	Notes and Assumptions
			author fees for individual chapters (covered separately if applicable).
Contributing Author Fees (if applicable)	\$0	\$15,000	Low: all chapters written by lead author; High: up to 4 contributing authors for specialty chapters (emerging tech, accessibility, networking, design). Per-chapter rates typically \$1,500–\$3,500 for textbook-quality writing.
Copyediting (300+ page manuscript + 5 companion documents)	\$8,000	\$15,000	Assumes professional educational copyeditor with AV or STEM content experience. Includes two passes: initial developmental copyedit after SME review and final proofread before production. Rate: approximately \$0.015–\$0.025 per word for educational technical content.
Graphic Design and Document Layout (Typesetting)	\$10,000	\$25,000	Includes: professional InDesign typesetting of textbook; slide deck design (12 chapter decks, consistent ISTE/AVIXA co-branded template); original diagram and signal flow illustrations; cover design. Low: uses institutional design resources; High: full external design agency engagement.
Video Production (40	\$40,000	\$120,000	This is the highest-

Budget Category	Low Estimate	High Estimate	Notes and Assumptions
videos, 3–7 min each; approx. 225 min total)			variance budget line item. Low: produced with institutional media center resources, student production team, and borrowed equipment; High: full professional video production crew with multi-camera setup, professional editing, motion graphics, and color grading. Mid-range (\$60,000–\$80,000) reflects a small professional production company using hybrid in-house/outsourced model.
Caption and Audio Description Production (40 videos)	\$4,000	\$8,000	Professional captioning service (not ASR only) for all 40 videos; audio description for the 22 videos requiring it. Rate: approximately \$150–\$300 per video for caption + transcript; \$200–\$400 per video for audio description. Budget assumes human-reviewed captions meeting DCMP (Described and Captioned Media Program) quality standards (98%+ accuracy).
LMS Assessment Package – QTI Build and Testing	\$3,000	\$8,000	Includes QTI 2.1 coding of all 120 questions with metadata; package structure and manifest file development; import testing in Canvas, Blackboard, Moodle, and D2L

Budget Category	Low Estimate	High Estimate	Notes and Assumptions
			Brightspace; bug-fixing and retest cycle. Low: performed by instructional technology staff; High: contracted to e-learning development vendor.
Accessibility Audit (documents + video)	\$3,000	\$6,000	WCAG 2.1 AA conformance audit of all six textbook documents in all output formats (DOCX, PDF/UA, PPTX, HTML); screen reader testing (JAWS/NVDA + VoiceOver); document structure remediation for PDF/UA compliance. Excludes video caption review (included in video caption line item above).
Project Management (8-12 weeks active, plus pilot phase)	\$8,000	\$15,000	Assumes a dedicated part-time project manager shared between ISTE and AVIXA. Includes: kickoff facilitation, weekly check-ins, SME coordination, milestone tracking, feedback consolidation, file management, and delivery coordination. Low: managed by internal ISTE or AVIXA staff member; High: contracted professional PM with publishing experience.
Legal Review (co-branding)	\$2,000	\$5,000	Review of co-branding agreement

Budget Category	Low Estimate	High Estimate	Notes and Assumptions
agreement, licensing, content accuracy)			between ISTE and AVIXA; Creative Commons license language; FERPA and accessibility legal content accuracy review. Each organization should engage their own legal counsel; budget reflects ISTE or AVIXA legal cost only (not combined). Low: handled by in-house counsel; High: outside counsel engagement.
Pilot Delivery Support (survey tools, stipends, travel)	\$1,000	\$4,000	Includes: survey tool subscription or setup; pilot institution faculty stipend (\$500 recommended); travel for on-site pilot support (if applicable). Low: fully remote pilot with no travel.
Miscellaneous (printing review copies, cloud storage, software licenses, admin)	\$2,000	\$5,000	Includes: printed review drafts for SMEs who prefer physical copies; cloud collaboration workspace subscription; InDesign and Adobe Acrobat licenses; miscellaneous administrative expenses.
TOTAL ESTIMATED RANGE	\$114,000	\$294,000	The single most significant cost driver is the video production model. Institutions choosing to produce all 40 videos in-house using existing

Budget Category	Low Estimate	High Estimate	Notes and Assumptions
			media center resources can reduce the total budget by \$60,000-\$80,000. ISTE and AVIXA are encouraged to explore co-production partnerships with member institutions, university media centers, or AVIXA corporate member companies with video production capabilities as a cost-reduction strategy.

Revenue and Cost Recovery Considerations

This budget represents the upfront development investment. Revenue and cost recovery pathways for ISTE and AVIXA to consider include: (1) **Institutional licensing:** Per-institution or per-program licensing for the complete package (textbook + all companion resources); (2) **Professional development delivery:** Fee-based instructor-led workshops using this curriculum, delivered by ISTE/AVIXA or licensed facilitators; (3) **Corporate member co-sponsorship:** AVIXA corporate members may be interested in co-sponsoring video production in exchange for acknowledgment (without commercial endorsement of products); (4) **Grant funding:** IMLS, NSF, Department of Education, and state workforce development agencies have funded open educational resource (OER) development in STEM and CTE fields; the CC BY-NC-SA license is compatible with most OER grant requirements.

Section J: Licensing, Rights, and Legal Notices

Recommended License

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License Element	Requirement	Application to This Work
Attribution (BY)	Users must give appropriate credit, provide a link to the license, and indicate if changes were made	Required citation format: <i>"AV Fundamentals for Educators," ISTE and AVIXA, 2026. Licensed under CC BY-NC-SA 4.0.</i> Attribution must appear on all copies, adaptations, and derivative works, including excerpts used in other instructional materials.
NonCommercial (NC)	Users may not use the material for commercial purposes	Institutions may reproduce, adapt, and distribute this material for non-commercial educational and training purposes without a separate license. Commercial use—including sale, commercial training programs where tuition is charged, and integration into commercial products—requires a separate licensing agreement with ISTE and AVIXA.
ShareAlike (SA)	Adaptations must be distributed under the same or compatible license	Any institution that adapts, localizes, translates, or builds upon this material

License Element	Requirement	Application to This Work
		must release its adaptation under CC BY-NC-SA 4.0 or a compatible Creative Commons license. Adapted versions must clearly identify their changes from the original.
Commercial Adoption	Separate licensing agreement required for commercial use	Institutions wishing to use this material for commercial purposes (fee-based professional development, corporate training, bundled sale with other commercial products) must contact ISTE and AVIXA for licensing terms and fees. Contact: [licensing@placeholder.org]

Co-Branding Guidelines

The following guidelines are placeholder specifications pending execution of the formal ISTE/AVIXA co-branding agreement. Final co-branding standards must be reviewed and approved by the communications and legal teams of both organizations before any materials are published or publicly distributed.

Element	Guideline (Placeholder)
Logo Placement – Cover	ISTE and AVIXA logos to appear together on the textbook front cover and all companion document title pages. Placement, sizing, and clear space requirements per each organization's current brand standards. Neither logo shall appear larger than the other.
Logo Placement – Interior	ISTE and AVIXA logos to appear on the copyright/legal page (verso of title page) and on the footer or header of the first page of each major companion document. Interior body pages do not require logo repetition.
Wordmark Usage	"ISTE" and "AVIXA" wordmarks and any associated descriptive text (e.g., "International Society for Technology in Education" and "the Audiovisual and Integrated Experience Association") to appear on the copyright page with correct

Element	Guideline (Placeholder)
	legal entity names as confirmed by each organization's legal team.
Digital Versions	All digital file formats (PDF, DOCX, PPTX, HTML, video opening/closing frames) to include both logos consistent with cover placement guidelines. Video content to include co-branded title card at opening and closing.
Legal Review Requirement	The final co-branding agreement must be reviewed and executed by legal counsel of both ISTE and AVIXA before any materials bearing both logos are published, distributed, or publicly displayed. This placeholder text does not constitute a binding co-branding agreement.

Vendor-Neutral Commitment Statement

Vendor-Neutral Commitment

All instructional content in the *AV Fundamentals for Educators* textbook package is intentionally and consistently vendor-agnostic. Product brand names, model numbers, manufacturer-specific terminology, and vendor-specific software commands do not appear in the instructional body text of the textbook, Instructor Guide, Lab Workbook, or this Resource Appendix. Specifications in the Bill of Materials templates and site survey forms describe required performance characteristics only.

Instructor supplemental notes and individual lab exercises may reference specific free and open-source software tools by name (e.g., draw.io for signal flow diagrams; Angry IP Scanner for network scanning) where a specific tool is necessary for practical instruction and no fee is charged. Any such reference is made for educational accuracy and does not constitute commercial endorsement by ISTE or AVIXA.

References to standards bodies, professional organizations, and regulatory frameworks (AVIXA, ISTE, ANSI, IEC, ADA, WCAG, CAST, EDUCAUSE, ASHRAE) are made for educational accuracy and legal compliance purposes. These references

do not constitute commercial endorsement of any organization's products, services, or certification programs beyond the educational context of this textbook.

Any SME reviewer who identifies a brand name or model number in instructional content should flag it in the revision matrix as a "vendor-neutral violation" for immediate correction by the author.

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and AVIXA request (but do not require) notification of translations to support a community registry of localized versions.

FERPA Notice

FERPA Compliance Notice

No student personally identifiable information (PII) should be collected, stored, or transmitted through any AV system described in this textbook without full compliance with the Family Educational Rights and Privacy Act (FERPA, 20 U.S.C. § 1232g) and applicable state student privacy laws. This notice applies specifically to the following AV practices described in this textbook: lecture recording and archiving, live streaming of class sessions, video conferencing session recording, and digital signage systems that may capture student images.

Instructors and institutions should consult their institution's privacy officer and legal counsel before: (1) recording, streaming, or archiving class sessions that include identifiable student participation; (2) using AI-powered AV features (auto-framing cameras, occupancy sensors, voice activity detection) that may generate or transmit student behavioral data; (3) integrating AV system logs or analytics with student information systems. Written consent processes for recording should be reviewed by the institution's FERPA compliance officer.

Errata Process

Post-publication errors—including factual inaccuracies, outdated standards references, accessibility corrections, and typographical errors—may be submitted by readers, instructors, and SMEs to the editorial team. The following process governs error reporting and correction:

- **Submission:** Submit errata reports to [errata@placeholder.org] with the subject line "AVFE Errata - [Chapter or Section]" and the following information: (a) document name and version, (b) page number or section

reference, (c) description of the error, (d) proposed correction with supporting source or rationale.

- **Review:** All submitted errata are reviewed by the Lead Author and at least one SME within 30 business days of receipt. Confirmed errors are logged in the public errata register.
- **Public errata log:** A public-facing errata log is maintained on the ISTE/AVIXA joint resource webpage and updated quarterly. All confirmed errors and their corrections are listed with the version in which they will be resolved.
- **Incorporation into editions:** Confirmed errata are incorporated into subsequent digital file updates and printed edition revisions. Version 1.1 (target 6 months post-launch) will incorporate all errata confirmed during the pilot phase.

Version History

Version	Date	Description of Changes	Approved By
1.0	[Publication Date – TBD]	Initial release of complete textbook package: textbook manuscript, Instructor Guide, Lab Workbook, Practice Exam, Resource Appendix, and Slide Decks (12 chapters). Includes QTI assessment package and 40 demo videos.	ISTE Editorial Director + AVIXA Editorial Director
1.1	[Target: 6 months post-launch]	Post-pilot revision: incorporates faculty and student feedback from pilot institutions; addresses all confirmed errata from Version 1.0; minor updates to Chapter 12 (Emerging	TBD (Lead Author + PM)

Version	Date	Description of Changes	Approved By
		Technology) to reflect technology developments since initial publication.	
1.2	[Target: 18 months post-launch]	Standards update: incorporates any revisions to referenced standards (ANSI S12.60, WCAG, ADA accessibility guidance) issued after initial publication; updated BOM cost ranges based on current market data.	TBD
2.0	[Target: 2-year review cycle]	Major content update: full review of all 12 chapters; updated emerging technology chapter with current developments in AI-powered AV, XR, and AoIP; refreshed video content for changed technologies; updated AVIXA domain alignment if exam content outline has been revised.	TBD (Full SME review cycle required for Version 2.0)

Standards Currency Notice

The AV industry and the regulatory environment governing educational technology and accessibility are subject to ongoing change. Standards referenced in this textbook (including ANSI S12.60, IEC 60118-4, WCAG 2.1, and ADA technical guidance) reflect the versions current at time of publication. Instructors are encouraged to check for updated versions of referenced standards annually and to consult the errata log for guidance on changes affecting instructional content. AVIXA's TechComm (Technical Standards

Committee) publishes updates to AV-specific standards; ISTE publishes updates to the ISTE Standards on a periodic basis. Both organizations' websites are authoritative sources for current standards versions.

AV Fundamentals for Educators - Resource Appendix and Production Companion Guide

Co-published by ISTE (International Society for Technology in Education) and AVIXA (the Audiovisual and Integrated Experience Association)

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